

An exotic $S^2 \times S^2$ from a certificate-checked surface-bundle surgery manifold

John Clyde

VentiMath

`verification@ventimath.org`

Version 2.4.0 — 2 September 2026

<https://doi.org/10.5281/zenodo.22254457>

Abstract

We define a compact marked surface-bundle surgery manifold V_{aud} , prove that $\pi_1(V_{\text{aud}}) = 1$, and double it along an explicit involution σ_{aud} of its boundary. The closed manifold $Z_{\text{aud}} = V_{\text{aud}} \cup_{\sigma_{\text{aud}}} V_{\text{aud}}$ is simply connected and spin with intersection form H , hence homeomorphic to $S^2 \times S^2$ by Freedman; it is a closed genus-two surface bundle over a genus-two surface surgered along four Lagrangian tori, hence symplectic with symplectic Kodaira dimension at least zero by Auroux–Donaldson–Katzarkov, Liu, and Ho–Li; hence, since Li’s Kodaira dimension is an invariant of the oriented diffeomorphism type, it is not diffeomorphic to $S^2 \times S^2$. The final classification and non-diffeomorphism deductions use named external theorems whose hypotheses are discharged in the text. Simple connectivity itself rests on ten explicit relations in the torus complement together with the two product-filling words: the group they present is trivial, a four-line coset enumeration, and the geometric content of the paper is that those relations and filling words hold. Sealed derivation certificates derive the same relations from the triangulation and re-prove triviality by a second route; their automated replay assumes that at least one checker conforms to the mathematical specification, and the two shipped implementations have not received an independent human line-by-line audit. The construction was motivated by Wuebben’s proposed exotic $S^2 \times S^2$; whether Z_{aud} is his manifold is an attribution question, recorded clause by clause as checklist D1–D14 in the supplement, on which no theorem here depends.

2020 Mathematics Subject Classification. 57R55 (primary); 57K40, 57M05 (secondary).

1 Introduction

This paper constructs a closed symplectic four-manifold homeomorphic but not diffeomorphic to $S^2 \times S^2$. The construction has three layers. A compact marked surface-bundle surgery manifold V_{aud} is defined from explicit data (Definition 2.1) and proved simply connected: finite derivation certificates first prove that four explicit presentations are trivial, and a geometric argument then reconstructs the marked bundle, torus exterior, peripheral curves, and drilled transport relation, and identifies one fixed presentation with $\pi_1(V_{\text{aud}})$. Elementary topology and a separate framing theorem give the intrinsic homology, spin, fiber, section, and symplectic statements of Theorem 2.2. Finally V_{aud} is doubled along an involution σ_{aud} of its boundary

defined from the same data, and the double Z_{aud} is shown to be homeomorphic to $S^2 \times S^2$ and symplectic of nonnegative Kodaira dimension (Theorem 2.3). The other three presentations are algebraic robustness checks, not additional geometric manifolds.

The construction was motivated by the disputed fundamental-group step in Wuebben’s proposed exotic-manifold construction [1], which builds on the surgery scheme of Lidman and Piccirillo [4]. A contrary computation attributed to Lidman and Piccirillo has been reported, but no presentation or nontriviality witness for it is publicly available in the materials examined here. This paper does not adjudicate that dispute. Its theorems concern the separately defined V_{aud} and Z_{aud} ; whether these coincide with Wuebben’s V and his double is an attribution question, recorded as the open checklist D1–D14 in the supplement, and no theorem here has a clause of that checklist among its hypotheses.

Triviality is deliberately not used as evidence that a proposed source-to-presentation dictionary is correct. Wuebben’s public computations report trivial groups for all 72 cases in one nearby grid and classify 14 of 15 single-word perturbations as SILENT at the group-decision layer [2]; the separate 100-case meridian-shift scan in this project likewise reports only trivial groups. These experiments show that triviality is non-discriminating in this neighborhood. They do not support the source comparison. Instead, the identification of the fixed presentation with V_{aud} is proved by the geometric reconstruction in Sections 4–5; the transfer to Wuebben’s member is the attribution question of Section 8.

The logical architecture is summarized below. Solid arrows are proved in this article; the dashed arrow is the attribution question, which no theorem here uses.

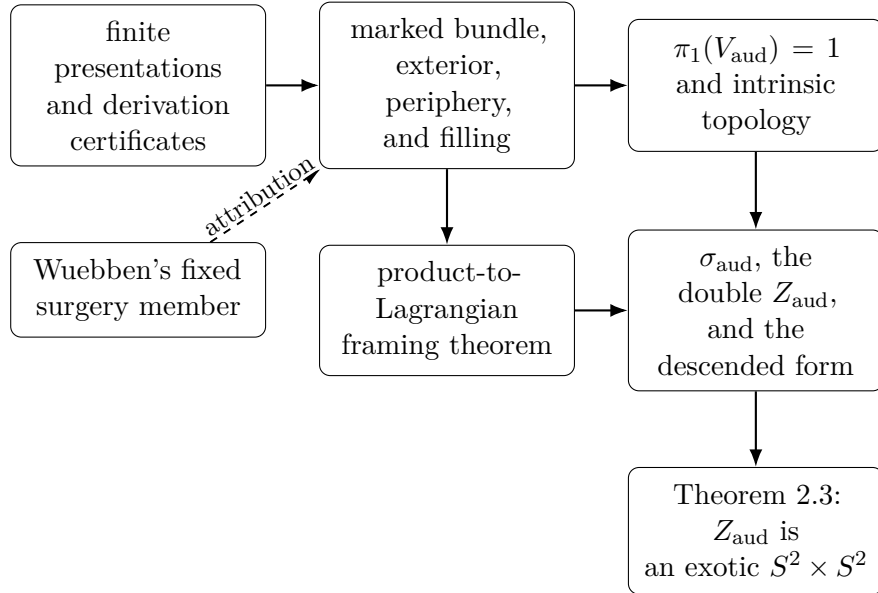


Figure 1: Dependency structure. Certificate checking, geometric conversion, symplectic framing, and doubling are different evidence layers; the dashed arrow is the attribution question, which no theorem uses.

The frozen transport and theorem-critical certificates replay in separate Python and Ruby implementations. Automated acceptance assumes that at least one checker conforms to the mathematical specification; neither has received an independent human line-by-line audit. The PL and symplectic arguments likewise await external specialist review. These review targets

are stated once, in Section 8; implementation inventories, the attribution checklist, and the historical transfer chain are kept in the companion supplement.

2 The audit manifold and principal statements

Write $\Sigma_{1,1}$ for an oriented once-punctured torus with based oriented spine loops α, β .

Definition 2.1 (the audit manifold). Let F be a closed oriented genus-two surface with an ordered five-chain (a, b, c, d, e) whose regular-neighborhood complement is the disjoint union of two disks containing marked points p, O . Let φ_0 be the orientation-preserving involution reversing the chain, fixing p, O , and acting freely on c by a half-rotation. Let

$$\psi_0 = T_a \circ T_b,$$

with T_b applied first and both twists supported away from a collar of e . Form the marked bundle R_{aud} over $\Sigma_{1,1}$ by clutching the α - and β -mapping cylinders of φ_0 and ψ_0 to one common marked fiber.

Inside R_{aud} put

$$T_\alpha = c \rtimes_{\varphi_0} S_\alpha^1, \quad T_\beta = e \times S_\beta^1.$$

Fix the basepoint $q = (p, *)$. Let $A = [\bar{\alpha}]^{-1}$ and $B = [\bar{\beta}]^{-1}$ be the based inverse base loops. Base the oriented meridians M, N along the initial fiber arcs y_1 to c_y and s_2 to s_e , respectively. In the product normal boundaries let λ_α be the y_1 -whiskered push-off that lifts the inverse α direction and closes along the half of c running from $c_s = \varphi_0(c_y)$ to c_y , oriented from c_s to c_y , and let λ_β be the s_2 -whiskered push-off of the inverse β direction along e . Their based words Ax and $(r^{-1}Mr)B$ are conclusions of Lemmas 5.11 and 5.12, not part of this definition. Define V_{aud} by the product-framed fillings

$$M\lambda_\alpha = 1, \quad N\lambda_\beta = 1.$$

The orientations of $M, N, \lambda_\alpha, \lambda_\beta$ are part of this definition. Thus this is the $(\epsilon_A, \epsilon_B) = (+1, +1)$ sheet in the fixed audit basis; changing one exponent while holding that basis fixed is not declared to preserve the filling.

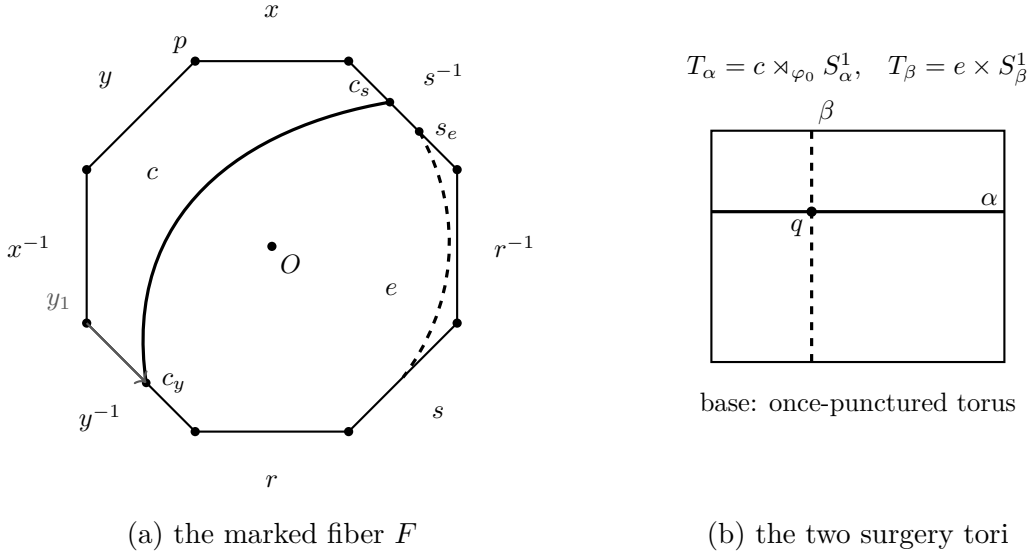


Figure 2: The audit manifold's marked data. Panel (a) is the symmetric octagon model of the genus-two fiber, with boundary word $xyx^{-1}y^{-1}rsr^{-1}s^{-1}$; all eight corners represent p , and O is the center. The marked curves c, e and the initial y_1 whisker determine the curve subbundles and peripheral basing. Panel (b) shows the two tori over the base loops. The drawing is explanatory; the proof uses the certified simplicial paths and incidences.

Theorem 2.2 (Audit-Manifold Theorem). The compact oriented smooth four-manifold V_{aud} has connected boundary, Euler characteristic 2, and

$$\pi_1(V_{\text{aud}}) = 1, \quad H_k(V_{\text{aud}}; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 0, 2, \\ 0, & k = 1, 3, 4. \end{cases}$$

It is spin. A genus-two fiber F disjoint from the surgery tori survives as a primitive square-zero class generating $H_2(V_{\text{aud}}; \mathbb{Z})$, and the fixed point p gives a properly embedded once-punctured-torus section Γ with $F \cdot \Gamma = 1$. Moreover V_{aud} admits a symplectic form for which F is symplectic.

Theorem 2.2 is source-independent: it concerns the explicit object in Definition 2.1. Simple connectivity is proved twice in Section 5: from ten explicit relations in the torus complement by a four-line coset enumeration (Section 5.1), and from the sealed certificates of Section 3 through the geometric identification at the end of Section 5; the remaining topological assertions are proved there, and symplecticity after the logically separate framing theorem in Section 6.

Theorem 2.3 (an exotic $S^2 \times S^2$). Let σ_{aud} be the involution $[z, t] \mapsto [\varphi_0(z), 1 - t]$ of the boundary mapping torus ∂V_{aud} (Lemma 7.1) and $Z_{\text{aud}} = V_{\text{aud}} \cup_{\sigma_{\text{aud}}} V_{\text{aud}}$. Then Z_{aud} is a closed symplectic four-manifold that is homeomorphic but not diffeomorphic to $S^2 \times S^2$.

Theorem 2.3 is proved in Section 7 from Theorem 2.2, the framing theorem, and published results whose hypotheses are discharged there one by one. It is a statement about the object defined here; its relation to Wuebben's proposed manifold is discussed in Section 8 and enters no proof.

3 The finite certificate theorem

Theorem 3.1 (finite-certificate statement). Let $\mathcal{M}$ be the manifest identified in the normative-root paragraph below. Its sealed-presentation entry binds a three-generator, seventy-eight-relator presentation Q and four ordered pairs of filling words indexed by $(\epsilon_A, \epsilon_B) \in \{(+, +), (+, -), (-, +), (-, -)\}$. Let $P_{\epsilon_A, \epsilon_B}$ be the presentation obtained from Q by adjoining the corresponding pair. Every one of these four explicitly manifest-bound presentations presents the trivial group.

Normative root. Every file this theorem depends on is bound by the v2.4.0 proof manifest $\mathcal{M}$, whose SHA-256 is

e8118b489a1b365c002a1931839fb419ab0f456aaecf724f2acf979612c9b5b9

and which lists the digest of each certificate, frozen input, and checker. The manifest path, sealed-input digest, git tree object, archive DOI, and literal replay commands are collected in the supplement. Verifying $\mathcal{M}$ and replaying its named certificates constitutes the complete operational replay of the finite artifact; its mathematical interpretation is fixed by Definitions 3.2–3.4 and Theorem 3.5.

Definition 3.2 (letters, words, and equation keys). Fix generators $g_1, \dots, g_n$. Source relators are signed words in $\{\pm 1, \dots, \pm n\}$, with j denoting g_j . Certificate records use the monoid alphabet $L_n = \{1, \dots, 2n\}$ and the decoding map

$$\text{dec}(2j - 1) = j, \quad \text{dec}(2j) = -j.$$

Thus the required inverse-letter table is $(2, 1, 4, 3, \dots, 2n, 2n - 1)$. If w is a signed word, write $\bar{w}$ for its reverse with all signs changed, $\text{fr}(w)$ for free reduction, and $\text{cr}(w)$ for the word obtained by additionally removing inverse first/last pairs until none remain. Define

$$\text{ck}(w) = \min_{\text{lex}} \{\text{cyclic rotations of } \text{cr}(w) \text{ and } \overline{\text{cr}(w)}\},$$

with $\text{ck}(1) = 1$. For certificate words u, v put

$$E(u, v) = \text{ck}(\text{dec}(u)\overline{\text{dec}(v)}).$$

Equality of equation keys therefore means equality after free and cyclic reduction, cyclic conjugacy, and inversion. Each operation preserves the normal closure of the equation word.

Definition 3.3 (literal rewrite trace). Suppose records $R_0, \dots, R_{i-1}$ have been accepted, with $R_r = (\ell_r, \rho_r)$. A trace on a word w is a list of pairs (r, p) with $r < i$ and $0 \leq p \leq |w|$. At that step the contiguous subword beginning at p must equal ℓ_r literally, and it is replaced by ρ_r . No cyclic matching, inversion, or implicit free reduction occurs inside a trace. Write $T(w)$ for its final word.

Definition 3.4 (certificate grammar). A `luttinger-kbmag-proof-v1` certificate is a finite ordered list of records (ℓ_i, ρ_i) , together with source-binding metadata and identity roots. Every side is a word over L_n . The case index must be an integer in the range of the source filling array. The source digest, case index and slug, generator count, full relator list, and inverse table must agree exactly with the hash-bound input. A record has exactly one of four proof forms.

- (i) An *inverse axiom* has $\rho_i = 1$ and $\ell_i = a \iota(a)$ for the required inverse table ι .
- (ii) An *input-relator record* names an in-range source relator q_k and satisfies $E(\ell_i, \rho_i) = \text{ck}(q_k)$.
- (iii) An *overlap record* names parents $a, b < i$, an integer offset d with $-|\ell_b| < d < |\ell_a|$, and two traces. The two parent left sides must agree literally on the resulting nonempty overlap. Let w be the shortest word containing both occurrences. Replacing the occurrence of ℓ_a by ρ_a gives one branch w_a , and replacing ℓ_b by ρ_b gives the other w_b . If the recorded traces give $u = T_a(w_a)$ and $v = T_b(w_b)$, validity requires $E(\ell_i, \rho_i) = E(u, v)$.
- (iv) A *change record* names an old record $o < i$ and traces on its two sides. Their literal outputs must equal the separately stored `reduced_left` and `reduced_right` words, say u, v , and validity requires $E(\ell_i, \rho_i) = E(u, v)$.

Finally the root array has length exactly $2n$; its entry for each $a \in L_n$ is an in-range record whose two sides are literally $(a, 1)$. Full-inventory mode also requires one correctly slugged certificate for each of the eight source fillings, with no repeated index or slug and with the stated presentation dimensions.

Theorem 3.5 (certificate soundness). Let G be the group defined by the source relators for a selected filling. If a certificate satisfies Definitions 3.2–3.4, then G is trivial.

Proof. Induct over the records. An inverse axiom is an equality in the free group. For an input record, the equation word is a freely and cyclically reduced cyclic conjugate or inverse of a defining relator, so its two sides agree in G . Every trace replaces a subword by the other side of an earlier valid equation and therefore preserves the represented element of G .

For an overlap, both untraced branches come from one source word by applying one earlier equality, so they agree in G ; their traces preserve that equality. Equality of equation keys transfers the resulting equality to (ℓ_i, ρ_i) . For a change, the two sides of an earlier valid equation are separately rewritten by earlier equations, and the same equation-key argument applies. Thus every accepted record is an equality in G . The root records say literally that every generator letter and its inverse equal 1, so $G = 1$. $\square$

Definitions 3.2–3.4 are the normative mathematical specification. The complete reference control flow and concrete file format are reproduced in the supplement and in `verification/luttinger/CERTIFICATE_SPEC.md`. The Python and Ruby checkers separately implement it using only their standard libraries. They share neither source code nor runtime, but both were produced within this project and have not received an independent human line-by-line audit.

Proof of Theorem 3.1. For each of the four selected entries, the certificate bundle prepared with this version contains a certificate bound to the manifest-listed source digest, case index, slug, complete relator list, generator count, and inverse table. The Python and Ruby implementations both accept every record and all six identity roots. The corresponding certificate inventory is recorded in the supplement. Hence Theorem 3.5 applies to each selected filling. Separately, two standalone checkers replay the frozen elementary Tietze transport from its serialized raw input to Q . No geometric interpretation of either byte string enters this proof. $\square$

Proposition 3.6 (human-readable terminal collapse). Write the three generators of the fixed $(+, +)$ presentation as a, b, c . Three accepted certificate equations imply the more readable consequences

$$a = bc, \quad ac^2 = 1, \quad b^2c = cab.$$

These three consequences already force $a = b = c = 1$.

Proof. Records 6649, 6662, and 6665 of the zero-based ancestry array are, literally,

$$ac^{-1} = b, \quad c^{-2} = a, \quad c^{-1}b^2 = abc^{-1}.$$

Right multiplication of the first by c , right multiplication of the second by c^2 , and left multiplication of the third by c followed by right multiplication by c give the three displayed consequences. Now $a = bc$ and $ac^2 = 1$ imply $bc^3 = 1$, so $b = c^{-3}$ and $a = c^{-2}$. Substitution into $b^2c = cab$ gives $c^{-5} = c^{-4}$, hence $c = 1$, and then $a = b = 1$. The certificate carries the long proof that the three short equations follow from the original eighty relators; this paragraph exposes the final algebraic reason the group dies. $\square$

The finite theorem is therefore relative only to its published mathematical specification and to the proposition that at least one implementation conforms to it. The rest of the paper asks the different question: what geometric object do the hash-bound presentations describe? The fixed $(+, +)$ presentation is identified with V_{aud} below; comparison with Wuebben is postponed to Section 8. The supplement records the complete implementation inventory and the detailed layer-by-layer trust boundary.

What the three generators mean. The three generators of Q are not privileged loops drawn in the surface-bundle picture. They are the survivors of a proof-producing Tietze transport that starts with one edge generator for each non-tree edge of the drilled simplicial exterior. The seventy-eight relators retain the full two-skeleton information after those eliminations. Geometric meaning is carried separately by eighty-nine tracked words: the original fiber and base loops, both meridians, both product longitudes, and the comparison paths are all expressed in the three survivors. Thus Q is a compact algebraic coordinate system for the torus exterior, while the tracked words are the dictionary back to its geometry. The proof below never assigns a geometric name to a survivor merely because it simplifies a relation.

4 Geometric identification of the audit manifold

This section proves that the $(+, +)$ finite presentation of Theorem 3.1 is a presentation of the explicitly defined V_{aud} . The other three sign sheets remain algebraic robustness checks. This section makes no comparison with the target paper's denotation; that logically separate problem is discussed in Section 8.

The genus-2 fiber is realized as a regular neighborhood of the specified five-chain of curves — five plumbed annular bands and two cone discs — with the hyperelliptic-type involution φ_0 realized simplicially; its fixed points, the curve actions $a \leftrightarrow e$, $b \leftrightarrow d$, and the free half-rotation of the middle curve c are executable assertions about the complex, not assumptions. The bundle R_{aud} is assembled from simplicial mapping cylinders over the once-punctured torus

base; the two surgery tori T_α, T_β arise as induced subcomplexes, and the assembled 4-complex passes manifold checks down to the vertex-link level.

The finite artifact inventory and checker acceptance results are kept in the supplement. They establish finite predicates only. The lemmas below supply the mathematical conversions from simplicial incidences, paths, and word lists to the stated topological conclusions; successful replay is not used as a substitute for those conversions.

The audit dictionary is displayed in Table 1; it is not reconstructed from the fact that the filled groups happen to collapse. Here “literal” means a stored edge path in the triangulation, with its basepoint and orientation retained.

Audit datum	Mathematical role	Finite realization
F, p, O	genus-two fiber with two fixed marks	closed simplicial surface; both fixed vertices pass surface- and link-level tests
a, b, c, d, e	ordered filling five-chain	independently reconstructed ribbon; oriented c at V_2 represents $(rx)^{-1}$
φ_0	$x \leftrightarrow r, y \leftrightarrow s$	simplicial half-turn fixing p, O , checked on based paths
$\psi_0 = T_a T_b$	$x \mapsto y^{-1}, y \mapsto yx, r \mapsto r, s \mapsto s$	ordered flip stack with T_b first; based monodromy replay
T_α	$c \rtimes_{\varphi_0} S_\alpha^1$	induced torus on the c -stack; local-flatness and subbundle checks
T_β	$e \times S_\beta^1$	induced product torus on the e -stack; local-flatness and product checks
A, B	$[\bar{\alpha}]^{-1}, [\bar{\beta}]^{-1}$	literal based edge paths from q , retaining their orientations
M, N	meridians based along y_1, s_2	oriented dual normal circles with the same literal whiskers; every normal fiber checked

Table 1: Marked-bundle dictionary for the fixed $n = 0$ member. Peripheral longitudes, which carry the sign-sensitive part of the identification, are displayed separately in Table 2.

Lemma 4.1 (equivariant marked fiber). There is an orientation-preserving homeomorphism $h : |F_K| \rightarrow F$ carrying the ordered curves (a, b, c, d, e) and marked points (p, O) to those of Definition 2.1 and satisfying $h\varphi_K = \varphi_0 h$. It carries the named based loops x, y, r, s with their displayed orientations.

Proof. Finite evidence. The independent ribbon reconstruction has

$$i(a, b) = i(b, c) = i(c, d) = i(d, e) = 1, \quad i(u, v) = 0 \quad \text{otherwise.}$$

Its regular neighborhood retracts to a graph with six vertices and ten edges, hence has Euler characteristic -4 . The two disjoint symplectic pairs (a, b) and (d, e) force genus two, so the neighborhood has two boundary components. The complement in the certified genus-two surface has Euler characteristic two and exactly those two boundary circles; it is therefore two disks. An exhaustive rotation-system check begins with the 36 cyclic orders at the independent four-valent crossings. Alternation, orientation preservation, two boundary faces, and preservation of the two faces leave exactly four systems; reversing the permitted curve

orientations normalizes all four to one equivariant oriented ribbon type. The primary and independent fibers realize that same labeled type, including the involution on every directed curve end and the two p/O faces.

Topological input. Map vertex disks and edge bands by the checked cyclic correspondence. These product maps agree on attaching intervals, so they give an equivariant homeomorphism of the ribbon neighborhoods. Each complementary disk is invariant: if the involution exchanged them it would have no fixed point there, contrary to the certified two-point fixed set. The periodic-disk classification [13, Theorem 3.1 and Proposition 3.2] identifies each nontrivial orientation-preserving order-two disk action with a half-turn and shows that a periodic disk map fixed on the boundary is the identity. More explicitly, quotient each complementary disk by its half-turn. Both quotients are disks with one marked branch point, and the already constructed equivariant boundary map descends to their boundary circles. The Alexander extension, chosen to carry branch point to branch point, lifts uniquely after one boundary lift is fixed; the lift is equivariant and agrees with the ribbon map on the boundary. This gives the required equivariant marked map relative to the boundary. The based path comparison then reads, literally,

$$\varphi_0 : (x, y, r, s) \mapsto (r, s, x, y).$$

No fundamental-group filling or downstream theorem enters this argument. $\square$

Lemma 4.2 (relative monodromy and graph clutching). The map h can be chosen so that it conjugates the two monodromy representatives relative to the full c and e collars. It consequently extends to an orientation-preserving fiberwise homeomorphism

$$H : |K| \longrightarrow (R_{\text{aud}})_{\text{top}}$$

carrying the basepoint, both torus subbundles, and their product collars to the corresponding marked objects.

Proof. Finite evidence. The literal based actions are

$$\varphi_0 : (x, y, r, s) \mapsto (r, s, x, y), \quad \psi_0 : (x, y, r, s) \mapsto (y^{-1}, yx, r, s).$$

The second action is obtained from complete whiskered paths, not free homotopy classes, and is replayed in two differently subdivided beta stacks. On the three-row c collar the alpha map is exactly the free shift by four columns. The beta representative is the ordered supported word $T_a \circ T_b$, with T_b first; every trace cell avoids the full e collar, and the restricted tetrahedra form exactly its product staircase.

Topological conversion. Both bundle models are the union of a common fiber block and two mapping cylinders over the rank-two spine of $\Sigma_{1,1}$; there is no base two-cell. For a monodromy f , write

$$M_f = (F \times [0, 1]) / ((z, 1) \sim (f(z), 0)).$$

If $hf_K = f_*h$, the formula $[z, t] \mapsto [h(z), t]$ respects the seam and has the analogous inverse. For an isotopic target representative, the standard interpolating formula $[z, t] \mapsto [k_t(z), t]$, with $f_*k_1 = k_0f_K$, gives the same conclusion. The alpha conjugacy is exact on the c collar. For beta, both sides use the identical supported twist word; interpolating twist profiles is relative to the disjoint e collar. The two handle maps restrict to h on their common fiber and therefore glue, as do their inverses. This constructs H directly; bundle classification and Dehn–Nielsen–Baer are independent controls, not proof inputs. $\square$

Corollary 4.3 (the marked torus pair). Under H , the induced subcomplexes $c \rtimes_{\varphi_K} S_\alpha^1$ and $e \times S_\beta^1$ map to T_α, T_β , together with their product collars and the p -section. All smooth and symplectic arguments may therefore be performed in the audit-defined smooth target R_{aud} ; no smoothing of $|K|$ is chosen.

The model results through Corollary 4.3 concern V_{aud} and use no source-comparison clause. Five structurally different constructions test subdivision, monodromy order, twist sign, labels, and orientations. These internal controls support the audit object; they do not identify it with the object denoted by a source diagram. That open comparison is discussed only in Section 8.

5 Peripheral data and the product filling

Let $T = T_\alpha \sqcup T_\beta$. Transport the explicit normal block neighborhood $N_{1/2}(T)$ below through the marked homeomorphism H , and use the resulting product-collared neighborhood as $\nu(T)$. Put $C_{\text{aud}} = R_{\text{aud}} \setminus \text{int } \nu(T)$. This section identifies the normal boundary of each component, its commonly based peripheral pair, and the two filling slopes. The statements are deliberately separated so that a failure of one cannot be hidden by triviality of the final groups. This section proves the intrinsic product-filling theorem; it uses no symplectic or Lagrangian-framing input.

5.1 The relation sheet and the collapse

Everything this section proves about $\pi_1(V_{\text{aud}})$ can be read from ten relations among eight based loops in C_{aud} . Base the bundle at a point q of $\Sigma_{1,1}$ off the spine loops α, β , so that the fiber over q misses both tori, and at the marked point p of that fiber. The eight loops are: the fiber loops x, y, r, s , the based rail parallels of a, b, e, d of Lemma 4.1, so that φ_0 carries x, y to r, s and $\psi_0 = T_a T_b$ acts by $x \mapsto y^{-1}$, $y \mapsto yx$, $r \mapsto r$, $s \mapsto s$; the base loops A, B , the lifts through the p -section (which misses both tori, since p lies on neither c nor e) of loops at q parallel to α and β , each disjoint from its own spine loop and crossing the other once; and the meridians M, N of T_α, T_β , whiskered from p along y_1 to $c_y = y \cap c$ and along s_2 to $s_e = s \cap e$ (Lemma 5.6, Figure 4). Among the fiber loops, y meets c once, at c_y ; s meets c once, at $c_s = \varphi_0(c_y)$, and e once, at s_e ; x and r meet neither c nor e (Figure 3). With the orientation sheet of the audit triangulation and $\delta = r^{-1}$:

Lemma 5.1 (the relation sheet). In $\pi_1(C_{\text{aud}})$,

$$\begin{aligned} Ax A^{-1} &= r, & Ay A^{-1} &= s, & Ar A^{-1} &= x, & As A^{-1} &= Ny, \\ Bx B^{-1} &= y^{-1}, & By B^{-1} &= M^{-1}yx, & Br B^{-1} &= r, & Bs B^{-1} &= (r^{-1} M^{-1} r)s, \\ [x, y][r, s] &= 1, & B(s^{-1} r^{-1} yx) B^{-1} &= r^{-1} s^{-1} x. \end{aligned}$$

Proof. The surface relation is the boundary of the fiber octagon (Lemma 4.1). Each transport relation $LgL^{-1} = w$ is the boundary of the membrane swept by the fiber loop g along the base loop L . In R_{aud} that boundary reads $LgL^{-1} = \varphi(g)$ with φ the monodromy of L ; in C_{aud} every point where the membrane meets a torus is punctured, and each puncture inserts that torus's meridian, whiskered back to p along the membrane, with the sign of the local intersection. The membrane of g along A lies over the loop parallel to α , so it misses T_α (which lies over α) and meets $T_\beta = e \times S_\beta^1$ only in the fiber over the one crossing with β , there exactly at the points of $g \cap e$; symmetrically the membrane of g along B misses T_β and meets T_α exactly at the points of $g \cap c$, in one fiber. Hence:

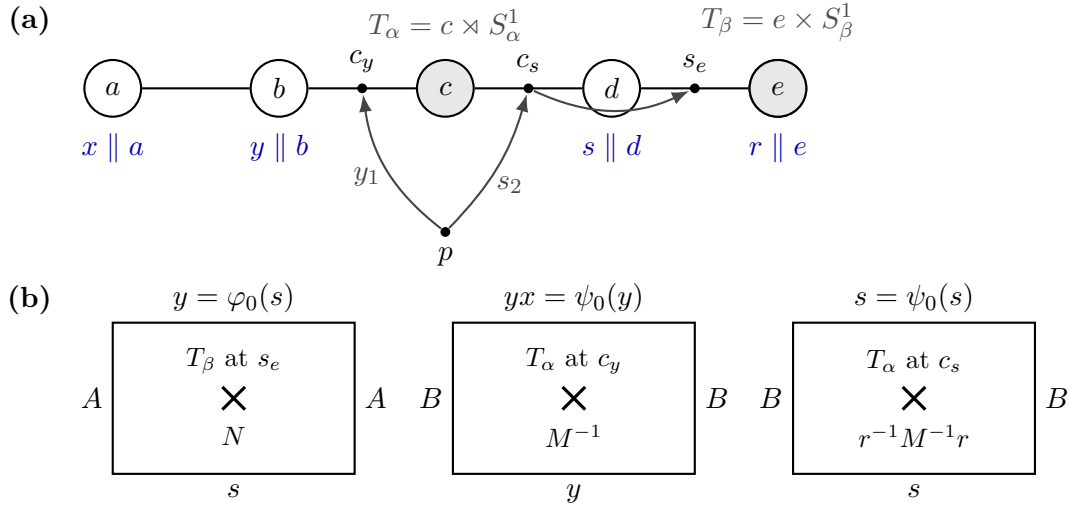


Figure 3: Where the meridians enter. (a) The intersection graph of the chain (a, b, c, d, e) : consecutive curves meet once and no others meet. The based loops x, y, s, r are rail parallels of a, b, d, e , so y meets c once (at c_y), s meets c once (at $c_s = \varphi_0(c_y)$) and e once (at s_e), and x, r meet neither c nor e , the fiber curves of the two tori. y_1 and s_2 are the whiskers that base M and N ; s_2 passes through c_s on its way to s_e . (b) The three transport membranes that meet a torus, each once: s along A meets T_β at s_e ($AsA^{-1} = Ny$); y along B meets T_α at c_y ($ByB^{-1} = M^{-1}yx$); s along B meets T_α at c_s ($BsB^{-1} = (r^{-1}M^{-1}r)s$). The basing square of λ_β meets T_α at the same point c_s through the same whisker (Figure 4). The five remaining transport membranes meet no torus. Schematic: basing and word order, not metric geometry.

- $AxA^{-1} = r$, $AyA^{-1} = s$, $ArA^{-1} = x$: x, y, r miss e , so the three membranes miss both tori and the relation is the action of φ_0 .
- $AsA^{-1} = Ny$: s meets e once, at s_e ; the membrane meets T_β there, and the meridian at s_e whiskered along s_2 is N itself.
- $BxB^{-1} = y^{-1}$, $BrB^{-1} = r$: x, r miss c ; the relation is the action of ψ_0 .
- $ByB^{-1} = M^{-1}yx$: y meets c once, at c_y ; the meridian there, whiskered along y_1 , is M (the same whisker that bases M), and the intersection sign gives M^{-1} .
- $BsB^{-1} = (r^{-1}M^{-1}r)s$: s meets c once, at c_s ; the meridian at c_s whiskered along the initial segment of s is $r^{-1}Mr$ in the basing of M (Lemma 5.12: the same crossing and whisker produce the conjugated meridian of λ_β), with sign -1 .
- The drilled-fiber relation is a consequence of the four B -transport relations and the surface relation: conjugating $s^{-1}r^{-1}yx$ by B letter by letter gives $(s^{-1}r^{-1}Mr)r^{-1}(M^{-1}yx)y^{-1} = s^{-1}r^{-1}xyy^{-1}$, and $s^{-1}r^{-1}xyy^{-1} = r^{-1}s^{-1}x$ is $[y, x] = [r, s]$, the surface relation (both steps are free-group identities, checked in `verification/luttinger/generation_check/`). Lemma 5.13 records the direct transport check.

The two meridian signs and the exact letter placement are certified, not inferred: each of the eight transport words, transported into the sealed three-generator presentation Q of the complement, reduces to the identity in a proof-producing completion replayed by both checkers (Section 3; supplement, certificate inventory). $\square$

Lemma 5.2 (generation). x, y, r, s, A, B, M, N generate $\pi_1(C_{\text{aud}})$.

Proof. This is not automatic: lifts of generators of $\pi_1(R_{\text{aud}})$ together with the meridians need not generate $\pi_1(C_{\text{aud}})$ (in a free group $\langle a, m \rangle$, the elements a^2ma^{-1} and m map onto the quotient by m yet generate a proper subgroup). It is a finite check. Lemma 5.7 presents $\pi_1(C_{\text{aud}})$ as the sealed three-generator, 78-relator presentation Q , and the sealed Tietze certificate transports each of the eight loops into Q as an explicit word. Coset enumeration of Q over the subgroup those eight words generate terminates with index 1 (`verification/luttinger/generation_check/`, replayed from the manifest-bound sealed file in under a second); so does enumeration over the six words x, y, r, s, A, B alone, as the relations $AsA^{-1} = Ny$ and $ByB^{-1} = M^{-1}yx$ predict. $\square$

Theorem 5.3 (the collapse). Let G be the group presented by the eight generators, the ten relations of Lemma 5.1, and the two product-filling relations

$$M \cdot Ax = 1, \quad N \cdot (r^{-1}Mr)B = 1.$$

Then $G = 1$. The same holds with any of the other three sign sheets (1) in place of $(+, +)$, and with the drilled-fiber relation omitted.

Proof. Coset enumeration. The transcript is `verification/luttinger/displayed_sheet/displayed_sheet.g`: the twelve words above, typed from this page, enumerate to a single coset in a quarter of a second (GAP 4.16; a few seconds per sheet in the pinned GAP 4.11.1 container of the recorded run, which is emulated), and so do the other three sheets and the four sheets without the drilled-fiber relation. The two one-filling controls in the same transcript have $H_1 \cong \mathbb{Z}$, so the enumeration is not collapsing everything it is handed. $\square$

Corollary 5.4 (simple connectivity from the sheet). If the product-framed longitudes of Definition 2.1 satisfy $\lambda_\alpha = Ax$ and $\lambda_\beta = (r^{-1}Mr)B$ in $\pi_1(C_{\text{aud}})$ (Lemmas 5.11 and 5.12), then $\pi_1(V_{\text{aud}}) = 1$.

Proof. $\pi_1(V_{\text{aud}}) = \pi_1(C_{\text{aud}})/\langle\langle M\lambda_\alpha, N\lambda_\beta \rangle\rangle$ by van Kampen. By Lemmas 5.2 and 5.1 and the two longitude identities, the eight loops generate it and satisfy the twelve relations of Theorem 5.3, so it is a quotient of $G = 1$. $\square$

Thus the geometric content of simple connectivity is Lemma 5.1, Lemma 5.2, and the two longitude identities; the group theory is Theorem 5.3. The sealed certificates of Theorem 3.1 reach the same conclusion by a second route that derives the presentation from the triangulation with no hand-written word: Lemma 5.7 presents $\pi_1(C_{\text{aud}})$ from the cellular structure, the Tietze certificate transports the eight loops into it, and the derivation certificates prove triviality without coset enumeration. Theorem 5.14 upgrades the quotient of Corollary 5.4 to an isomorphism.

Remark 5.5 (what the controls do and do not show). At unit slopes every filling of the sheet has $H_1 = 0$ (Proposition 5.8 with $|p| = |q| = 1$), all four sign sheets collapse (Theorem 5.3), and so does every one of the 100 meridian-shifted presentations of the framing scan; abelianization and triviality are therefore blind to the choice of sheet and to those shifts. Coset enumeration time is not evidence either: replacing λ_α by $\lambda_\alpha c^{-1}$ or λ_β by $\lambda_\beta r^{-1}$ (fiber-direction shifts of the filling slope) lengthens the enumeration from a fraction of a second to minutes and can

exceed a fixed coset ceiling, yet every such filling that has completed is again trivial. What Proposition 5.8 does show is that the complement has infinitely many fillings, at non-unit slopes, that are provably not simply connected. A unit-slope filling that is provably not simply connected has not been exhibited.

5.2 Normal frontier and compact exterior

Lemma 5.6 (local flatness and the normal frontier). The two torus subcomplexes are disjoint, full, and locally flat. The barycentric level set

$$N_{1/2}(T) = \left\{ x : \sum_{v \in T} \lambda_v(x) \geq \frac{1}{2} \right\}$$

is an explicit normal D^2 -block bundle, and its frontier $F_{1/2}(T)$ is the normal-circle boundary. The derived frontier used by the computation is PL homeomorphic to $F_{1/2}(T)$, carrying every extracted dual loop to a literal oriented normal-circle fiber.

Proof. For every simplex $\sigma \subset T$, the complete link pair is checked to be

$$(S^3, \text{unknotted } S^1), \quad (S^2, S^0), \quad (S^1, \emptyset)$$

in dimensions 0, 1, 2, respectively. The enumeration covers every simplex of both tori. Vertex-link 3-spheres and unknotted link circles carry independently replayed collapse and cyclic-knot witnesses. Since the star pair of a simplex is the join of that simplex with its link pair, these standard links give the required standard local chart at every point. This is the simplex-link formulation of PL local flatness; see [14, Chapter 4, pp. 50–51].

In a simplex with torus face A and opposite face B , barycentric coordinates give the unique block form

$$N_{1/2}(T) \cap \sigma \cong \Delta_A \times C(\Delta_B), \quad F_{1/2}(T) \cap \sigma \cong \Delta_A \times \Delta_B.$$

The formulas agree on common faces. If $b(s)$ is a mixed-simplex vertex of the computed derived frontier, the map

$$b(s) \mapsto \frac{1}{2}b(s \cap T) + \frac{1}{2}b(s \setminus T)$$

extends cellwise to the canonical subdivision map for $\Delta_A \times \Delta_B$ and hence gives the global PL homeomorphism. Every mixed vertex and derived comparability is checked. For every torus triangle the alternating dual loop has constant torus coordinate and is precisely the subdivided normal link circle. No regular-neighborhood uniqueness or smoothing theorem is used. $\square$

Lemma 5.7 (complement presentation). Let $L = K[V(K) \setminus V(T)]$ be the full subcomplex on the vertices outside T . There is a homotopy-equivalence diagram

$$|L| \hookrightarrow |K| \setminus \text{int } N_{1/2}(T) \hookrightarrow |K| \setminus |T|, \quad |K| \setminus \text{int } N_{1/2}(T) \cong C_{\text{aud}},$$

in which both spaces on the right strongly deformation retract onto $|L|$. Consequently the maximal-tree edge generators of L and its oriented triangle boundaries give a presentation of $\pi_1(C_{\text{aud}})$. A sealed Tietze certificate reduces the raw cellular presentation to the three-generator, seventy-eight-relator presentation Q of Theorem 3.1, while transporting all named peripheral and comparison words. Exact cellular and transport counts are in the supplement.

Proof. On each simplex not contained in the full subcomplex T , let $r(x)$ be the point obtained by setting the coordinates on T to zero and renormalizing the remaining barycentric coordinates. The straight-line homotopy $H_t(x) = (1-t)x + tr(x)$ agrees on faces and fixes $|L|$. If $\rho(x) = \sum_{v \in T} \lambda_v(x)$, then $\rho(H_t(x)) = (1-t)\rho(x)$. Thus H_t stays in $|K| \setminus |T|$ and also restricts to the block exterior $\{\rho \leq 1/2\} = |K| \setminus \text{int } N_{1/2}(T)$. The homeomorphism H and the chosen product collar identify that block exterior with C_{aud} . This proves all three asserted homotopy identifications, including the previously implicit passage from the deleted subcomplex to the compact torus exterior.

Collapsing a maximal tree and applying cellular van Kampen to every oriented two-simplex gives the raw presentation. Each recorded Tietze elimination is replayed against the unique current relator that implies it; separate Python and Ruby checker implementations agree on the final presentation and every tracked word. $\square$

Based meridians and product-framing longitudes for both tori are traced as literal simplicial loops carrying the audit definition's basing whiskers. External consistency checks compare the resulting words with Wuebben's corrected relation sheet and basing convention, but those comparisons play no logical role in any theorem about V_{aud} . Internally, a combinatorial sweep of the relevant transport square resolves the basing double-count question from the audit paths themselves. Both surgery tori are certified locally flat at every simplex link, and every extracted dual meridian is checked to be a literal normal-circle fiber of an explicit normal block model, with no regular-neighborhood theorem invoked. The doubled-section datum used in the historical target application is kept in the supplement; the intrinsic section of V_{aud} is treated directly below.

5.3 Product fillings and the slope family

The relation sheet of Lemma 5.1 is the coordinate presentation read from the literal paths of the audit triangulation, before filling. The two commonly based product-framed directions are

$$\lambda_\alpha = Ax, \quad \lambda_\beta = (r^{-1}Mr)B,$$

and the four algebraic sign sheets append exactly

$$M\lambda_\alpha^{\epsilon_A} = 1, \quad N\lambda_\beta^{\epsilon_B} = 1, \quad (\epsilon_A, \epsilon_B) \in \{\pm 1\}^2. \quad (1)$$

Proposition 5.8 (abelianization of a slope family). For $p, q \in \mathbb{Z}$, let $P_{p,q}$ be the quotient of the sealed complement presentation Q obtained by adjoining

$$M\lambda_\alpha^p = 1, \quad N\lambda_\beta^q = 1.$$

With the convention $\mathbb{Z}/0 = \mathbb{Z}$,

$$H_1(P_{p,q}; \mathbb{Z}) \cong \mathbb{Z}/q \oplus \mathbb{Z}/p.$$

In particular the complement has infinitely many fillings with nontrivial fundamental group. Omitting either filling relation leaves an infinite cyclic summand, while $|p| > 1$ or $|q| > 1$ produces nonzero torsion in abelianization.

Proof. Let g_1, g_2, g_3 be the generators of Q in its sealed order. The exponent vectors of its 78 relators span exactly $\mathbb{Z}(0, 0, 1) \subset \mathbb{Z}^3$, so

$$H_1(Q; \mathbb{Z}) = \mathbb{Z}\langle g_1 \rangle \oplus \mathbb{Z}\langle g_2 \rangle.$$

The hash-bound tracked words have exponent vectors

	M	N	λ_α	λ_β
vector in $\mathbb{Z}^3$	$(0, 0, 1)$	$(0, 0, 0)$	$(0, 1, 2)$	$(-1, 0, -3)$

Thus their classes in $H_1(Q)$ are respectively $0, 0, g_2, -g_1$. The two new relations impose $pg_2 = 0$ and $qg_1 = 0$, which gives the displayed answer. The single-filling and nonunit-coefficient assertions follow immediately. $\square$

These are four algebraic specializations in one fixed symbolic basis. Only the $(+, +)$ specialization is used below as a presentation of V_{aud} . The finite presentations of Theorem 3.1 are the frozen Tietze images of all four literal word lists; for the other three, no identification with a filled geometric manifold is asserted here. At unit slopes the proposition is silent ($H_1 = 0$ for all four sheets), and Remark 5.5 says what the controls do and do not show.

Lemma 5.9 (extension across a torus filling). Let $X = T^2 \times D^2$ and order an integral basis of its boundary as $(\mu, \gamma_1, \gamma_2)$, where μ bounds the disk fiber. A boundary diffeomorphism $f : \partial X \rightarrow \partial X'$ extends over X if and only if $f_*(\mu) = \epsilon\mu'$ for $\epsilon \in \{\pm 1\}$. In these bases its matrix then has the form

$$[f_*] = \begin{pmatrix} \epsilon & u_1 & u_2 \\ 0 & a_{11} & a_{12} \\ 0 & a_{21} & a_{22} \end{pmatrix} = \begin{pmatrix} \epsilon & u \\ 0 & A \end{pmatrix}, \quad A \in GL(2, \mathbb{Z}), \quad u \in \mathbb{Z}^2.$$

Consequently a diffeomorphism of torus exteriors that carries each killed attaching slope to the corresponding slope up to sign extends across the filled pieces; the two surviving boundary directions are recorded by A and u .

Proof. Necessity follows because the kernel of $H_1(\partial X) \rightarrow H_1(X)$ is the primitive subgroup generated by μ . For sufficiency, write $x \in \mathbb{R}^2/\mathbb{Z}^2$ for the base coordinate and $z \in D^2 \subset \mathbb{C}$ for the disk coordinate. Up to translations, the displayed matrix is realized by

$$(x, z) \mapsto \begin{cases} (Ax, e^{2\pi i \langle u, x \rangle} z), & \epsilon = +1, \\ (Ax, e^{2\pi i \langle u, x \rangle} \bar{z}), & \epsilon = -1. \end{cases}$$

This restricts to the required affine isotopy class on the boundary. The linearization theorem for the flat three-torus [15, case (IV), pp. 464–465] says that every diffeomorphism of T^3 is isotopic to an affine representative; its linear part is the induced action on integral first homology. Extending that isotopy through a boundary collar adjusts the displayed extension to f .

The orientation sign of the displayed four-dimensional map is $\epsilon \det A$, which is also $\det[f_*]$ on the ordered boundary basis. Thus an orientation-preserving boundary map has an orientation-preserving extension, and an orientation-reversing boundary map has an orientation-reversing extension. In the filled-exterior application one conjugates the exterior

boundary map by the two attaching maps. If both attaching maps reverse the boundary orientation, their two signs cancel, so the conjugated map has the same orientation sign as the exterior map. This reduces the extension problem to the same meridian criterion and accounts explicitly for the third boundary direction as well as the orientation. $\square$

Corollary 5.10 (peripheral convention changes). Let (μ, λ, ν) be an oriented integral basis of the boundary three-torus of one deleted torus, with geometric filling slope $s = \mu + \lambda$. Reversing the orientations used to name both μ and λ , while leaving ν fixed, is the coordinate change

$$D_- = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in SL(3, \mathbb{Z}).$$

It carries s to $-s$ and hence preserves the unoriented filling slope. The corresponding boundary map extends across the filling $T^2 \times D^2$. In the commonly whiskered peripheral subgroup, the same relation is written

$$(\mu\lambda)^{-1} = \lambda^{-1}\mu^{-1} = \mu^{-1}\lambda^{-1} = 1,$$

where the second equality uses commutativity of the peripheral $\pi_1(T^3)$. By contrast, reversing only the displayed longitude changes the slope to $\mu - \lambda$; the matrix $\text{diag}(1, -1, -1) \in SL(3, \mathbb{Z})$ sends s to that distinct primitive class. It is not a convention change preserving the fixed filling unless an additional self-diffeomorphism of the complement carrying $\mu + \lambda$ to $\mu - \lambda$ is supplied.

Proof. The two displayed matrices give the stated changes on $H_1(T^3; \mathbb{Z})$. Since s and $-s$ define the same primitive kernel, Lemma 5.9 extends the first coordinate change over the filling piece. Because the meridian and longitude use a common whisker, their boundary subgroup is abelian; hence inversion of $\mu\lambda$ gives the displayed word. Finally $(1, -1, 0)$ is not $\pm(1, 1, 0)$, proving the last assertion. The same argument applies independently on the alpha and beta boundary components. $\square$

Direction	Based word	Geometric construction and evidence
λ_α	Ax	y_1 -based product push-off at c_y ; literal path construction and complement-only certificate
λ_β	$(r^{-1}Mr)B$	s_2 -based product push-off at s_e ; punctured transport square and complement-only certificate

Table 2: The two sign-sensitive longitude identifications. Neither certificate contains a filling relation. The words are oriented in the fixed audit basis; Lemma 5.10 states the complete coordinate change required to preserve the filling.

5.4 Peripheral curves and drilled transport

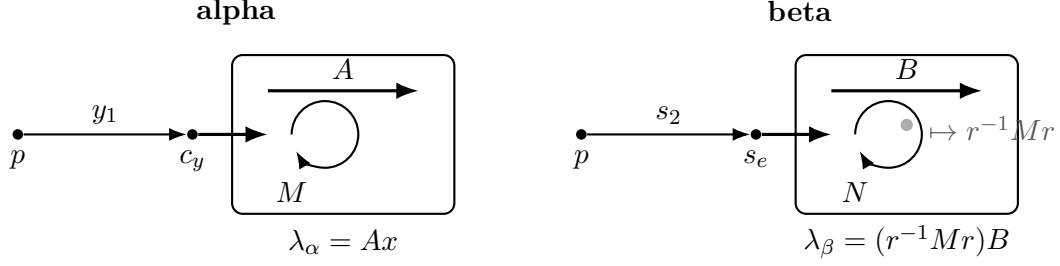


Figure 4: Common-whisker peripheral construction. On the alpha side the meridian and longitude use the same y_1 path to c_y . On the beta side they use the same s_2 path to s_e ; the swept basing square meets T_α once, and puncturing that intersection contributes the conjugated meridian $r^{-1}Mr$. The diagram records basing and word order, not metric geometry.

Lemma 5.11 (the alpha peripheral pair). Under the normal-frontier identification of Lemma 5.6, the extracted loops are the commonly based meridian and product longitude (M, λ_α) , and

$$\lambda_\alpha = Ax \quad \text{in } \pi_1(C_{\text{aud}}).$$

The displayed equality is certified using only the 78 complement relators, with neither filling relation present.

Proof. Consider T_α . The curve c meets the oriented y edge once, at c_y . Starting at p , the stored path follows the initial segment y_1 to c_y , makes a normal jog to the frontier of the explicit normal block, and traverses the oriented dual normal circle. Returning along the same whisker gives M . The longitude uses that identical whisker and normal jog, traverses the base direction A , and closes along that same half of c , from $c_s = \varphi_0(c_y)$ to c_y . These two literal based paths are what M and λ_α denote throughout and what the finite presentation fills with.

That the two loops use the same whisker is not an inference: in the sealed transport input they are $y_1\mu_\alpha y_1^{-1}$ and $y_1\lambda_\alpha^\circ y_1^{-1}$ with the identical whisker y_1 . So no independent conjugation is hidden in the filling word $M\lambda_\alpha^{\epsilon_A}$.

The identification of the extracted longitude with Ax has two independent supports. Geometrically it follows from how the path is built — based on the y rail at the unique crossing c_y , whiskered by y_1 , jogged to the normal-block frontier, lifting the base generator A , and closed along the half of c from c_s to c_y . Algebraically, the elementary elimination pass over the sealed presentation leaves a nonempty residual for $\lambda_\alpha^{-1}Ax$, but a proof-producing completion reduces that exact word to the identity. Independent Python and Ruby verifiers replay its complete ancestry cone from the sealed three-generator, 78-relator complement; neither filling relation is present. The supplement gives the exact path identifiers, word lengths, certificate inventory, and frozen source locations. $\square$

Lemma 5.12 (the beta peripheral pair). Under the same normal-frontier identification, the extracted loops are the commonly based meridian and product longitude (N, λ_β) , and

$$\lambda_\beta = (r^{-1}Mr)B \quad \text{in } \pi_1(C_{\text{aud}}).$$

Again the equality is certified in the unfilled complement.

Proof. The beta-coordinate word $\lambda_\beta^{-1}(r^{-1}Mr)B$ likewise leaves a nonempty residual under elementary elimination. The same proof-producing complement system reduces that exact

word to the identity; separate Python and Ruby verifier implementations replay its complete ancestry cone. Thus both displayed coordinate identities are algebraically certified in the complement, although Theorem 3.1 remains more direct: it fills with the literal stored boundary longitudes rather than relying on either coordinate substitution.

The beta extraction is the sign-sensitive case. The corresponding path uses the initial segment s_2 to the unique crossing s_e , the base loop B , and the fiber-normal push-off. Its swept basing square crosses T_α once. Puncturing that square contributes the conjugated meridian $r^{-1}Mr$, which is why the audit word for this direction is $(r^{-1}Mr)B$ and not B . The extracted paths are exported without coordinate substitution; they are what N and λ_β denote, and the identification with $(r^{-1}Mr)B$ is certified in the complement exactly as in the α case. An independently written extractor recovers both pairs without importing the complement, sweep, or peripheral modules. Local link checks certify both tori at every simplex, and the explicit block-frontier map checks every dual loop as a normal-circle fiber; these are the finite facts behind the geometric words “meridian” and “push-off.” $\square$

Lemma 5.13 (the drilled-fiber relation). The based relation

$$B(s^{-1}r^{-1}yx)B^{-1} = r^{-1}s^{-1}x$$

holds in $\pi_1(C_{\text{aud}})$. It is a formal consequence of the four B -transport relations and the surface relation (Lemma 5.1), so it adds nothing to the presentation; it is kept as an independent check of the transport machinery, because it is also verified directly.

Proof. The derivation is in the proof of Lemma 5.1: conjugating $s^{-1}r^{-1}yx$ by B letter by letter, the two meridian factors cancel and the residual is a conjugate of the surface relator. Independently, the transported loop κ_3 misses c but meets e once, so sweeping it along the core of the beta curve would meet T_β ; the audit sweeps it along the specified parallel push-off instead, and along that push-off the complete mapping-cylinder stack has no vertex on either surgery torus. A simplicial homotopy grid at p joins the push-off basepoint loop to the core one and identifies it with the presentation’s whiskered generator; it too is disjoint from both tori, and the based monodromy target reduces to $r^{-1}s^{-1}x$. The finite checker exhausts the stack, both torus vertex sets, every grid cell, and the target path, so the entire transport homotopy lies in C_{aud} ; Theorem 5.3 does not need it. $\square$

5.5 Simple connectivity of the product-filled audit model

Theorem 5.14 (audit-model identification). For the fixed audit sheet $(\epsilon_A, \epsilon_B) = (+1, +1)$ the natural map is an isomorphism

$$P_{+,+} \cong \pi_1(V_{\text{aud}}).$$

No geometric identification of $P_{+,-}, P_{-,+}, P_{-,-}$ is part of this statement. No clause of the open source-comparison checklist D1–D14 enters this theorem.

Proof. Lemmas 4.1 and 4.2 construct the marked bundle homeomorphism and carry the two curve subbundles and their product collars to the specified tori. Lemma 5.6 identifies their normal boundaries and meridians. Lemmas 5.11 and 5.12 identify the two commonly based product longitudes, including the conjugated meridian contributed by the punctured beta sweep. Lemma 5.13 places the final transport relation in the drilled complement.

By Definition 2.1, attaching the two product-framed filling pieces adds exactly the normal closure

$$\langle\langle M\lambda_\alpha, N\lambda_\beta \rangle\rangle.$$

Cellular van Kampen therefore gives a surjection from the filled presentation. Lemma 5.7 gives an actual presentation of the complement, not merely a list of valid relations; applying van Kampen with the two filling relations upgrades that surjection to the asserted isomorphism. Changing a common whisker simultaneously conjugates both elements of a peripheral pair and does not change their normal closure. Lemma 5.10 records the different, simultaneous meridian–longitude reversal that preserves a filling slope. This completes the identification with Definition 2.1 without using a symplectic form, a Weinstein chart, or any comparison with Wuebben’s text and figures. $\square$

Proof of Theorem 2.2, except symplecticity. Use the $(+, +)$ sheet in Theorem 5.14. Its source is trivial by Theorem 3.1; a quotient of the trivial group is trivial. Notice that this conclusion needs only surjectivity in Theorem 5.14, no symplectic or Lagrangian-framing result, and no source-identification assumption.

The boundary of R_{aud} is the F -bundle over $\partial\Sigma_{1,1} \cong S^1$ whose monodromy is the commutator determined by the oriented boundary word in the two clutching maps. It is connected. Both torus surgeries take place in the interior, so this is also ∂V_{aud} . Since

$$\chi(F) = 2 - 2 \cdot 2 = -2, \quad \chi(\Sigma_{1,1}) = 2 - 2 - 1 = -1,$$

multiplicativity for the bundle gives $\chi(R_{\text{aud}}) = 2$. Removing and regluing two copies of $T^2 \times D^2$, each with Euler characteristic zero and boundary T^3 , does not change Euler characteristic. Hence $\chi(V_{\text{aud}}) = 2$.

Simple connectivity gives $H_1(V_{\text{aud}}; \mathbb{Z}) = 0$. Because both the manifold and its boundary are connected, the beginning of the cohomology sequence of the pair, together with $H^1(V_{\text{aud}}; \mathbb{Z}) = 0$, gives $H^1(V_{\text{aud}}, \partial V_{\text{aud}}; \mathbb{Z}) = 0$. Lefschetz duality therefore gives $H_3(V_{\text{aud}}; \mathbb{Z}) = 0$. Similarly the homology sequence of the pair gives $H_1(V_{\text{aud}}, \partial V_{\text{aud}}; \mathbb{Z}) = 0$; duality and the universal coefficient theorem then show that $H_2(V_{\text{aud}}; \mathbb{Z})$ is torsion-free. These are the cases of the universal coefficient theorem and Lefschetz duality stated in [16, Theorems 3.2 and 3.43]. Since the boundary is nonempty, $H_4 = 0$, and the Euler characteristic now forces $H_2 \cong \mathbb{Z}$.

Choose a base point in the interior of the base away from the two clutching loops. Its fiber F misses both surgery tori, survives the fillings, and has trivial normal bundle, hence $F^2 = 0$. The fixed point p , which lies on neither c nor e , gives a proper section $\Gamma \cong \Sigma_{1,1}$ disjoint from the surgery tori. It also survives, and $F \cdot \Gamma = 1$. Thus $[F]$ is primitive and generates H_2 . Modulo two, H_2 is generated by $[F]$. On F the oriented splitting $TV_{\text{aud}}|_F = TF \oplus \nu F$, the Whitney product formula, and the mod-two reduction of the two Euler classes give

$$\langle w_2(V_{\text{aud}}), [F] \rangle = \chi(F) + F^2 = -2 + 0 = 0 \pmod{2}.$$

See [17, §§4 and 9]. Since evaluation on $[F]$ detects $H^2(V_{\text{aud}}; \mathbb{Z}/2)$, it follows that $w_2(V_{\text{aud}}) = 0$, so V_{aud} is spin. The remaining symplectic assertion is proved after Theorem 6.3. $\square$

6 The product-to-Lagrangian framing bridge

The proof of simple connectivity and the intrinsic topological assertions is complete before this section begins. The framing theorem is not used in Theorem 5.14. It proves the

source-independent symplectic assertion of Theorem 2.2; it is also the relevant bridge in any future comparison with a Lagrangian source construction.

Lemma 6.1 (bundle symplectic form and protected collars). The monodromy representatives in Definition 2.1 may be chosen to preserve one positive area form Ω on F , with ψ_0 equal to the identity on a collar of e and φ_0 equal to the stated half-rotation on a collar of c . Write $\pi : R_{\text{aud}} \rightarrow \Sigma_{1,1}$ for the bundle projection. If η is a positive area form on $\Sigma_{1,1}$, then for every $K > 0$

$$\omega_K = \Omega_{\text{vert}} + K\pi^*\eta$$

is a closed symplectic form on R_{aud} . The two tori T_α, T_β are Lagrangian. On disjoint protected neighborhoods of them there are oriented coordinates $(\theta_1, t, \theta_2, u)$ in which

$$\omega_K = f(\theta_1, t) dt \wedge d\theta_1 + K du \wedge d\theta_2, \quad f > 0,$$

with no mixed vertical–horizontal terms. On the alpha neighborhood these coordinates obey the seam $(\theta_1, t, 2\pi, u) \sim (\theta_1 + \pi, t, 0, u)$; on the beta neighborhood they are ordinary product coordinates.

Proof. Average any positive area form over the order-two action of φ_0 . An equivariant tubular coordinate on the invariant curve c then writes $\varphi_0(\theta, t) = (\theta + \pi, t)$ and the averaged form as $f(\theta, t)dt \wedge d\theta$. For this same area form, represent T_a and T_b by the usual area-preserving annular shears. Because e is disjoint from a and b , their supports may be chosen disjoint from a protected collar of e ; hence $\psi_0 = T_a \circ T_b$ preserves Ω and is the identity on that collar. Thus both clutching maps preserve one area form and retain the required collar behavior.

Use the flat product atlas determined by the two locally constant clutching maps. The fiber form descends in that atlas to a globally defined closed vertical form Ω_{vert} : on overlaps the transition is locally constant in the base and preserves Ω . Therefore $d\omega_K = 0$, and, with the fiber and base orientations fixed by Ω and η ,

$$\omega_K^2 = 2K \Omega_{\text{vert}} \wedge \pi^*\eta > 0.$$

This proves symplecticity and fixes the total orientation. Each surgery torus has one fiber tangent and one base tangent, so both summands restrict to zero on it.

Choose $\eta = du \wedge d\theta_2$ on disjoint annular collars of the two base loops and write $\Omega = f(\theta_1, t)dt \wedge d\theta_1$ on the protected fiber collars. The flat atlas then gives the displayed split expressions; there are no mixed terms to remove. The identity behavior of ψ_0 gives the beta product chart, while the half-rotation of φ_0 gives exactly the alpha seam. These choices can be made simultaneously because the protected collars are disjoint. We orient the normal momentum plane by (t, Ku) ; its boundary orientation is the positive meridian convention used below. $\square$

Definition 6.2 (Lagrangian-framing class). Let T be either audit torus with its oriented product coordinates, and let $\gamma \subset T$ be one of the named primitive curves. In a Weinstein chart fixing the zero section, take a sufficiently small nonzero constant covector copy of γ and radially project it to $\partial\nu(T)$. Its isotopy class in $\nu(T) \setminus T$ is the chartwise Lagrangian push-off class. The chart-independence part of Theorem 6.3 proves that this class is independent of the Weinstein chart and of the sufficiently small constant covector; the resulting boundary class is what we call the canonical Lagrangian-framing class of γ .

Theorem 6.3 (product framing equals Lagrangian framing). For the symplectic form ω_K of Lemma 6.1, the product-framed push-offs in Lemmas 5.11 and 5.12 represent the canonical Lagrangian-framing classes of T_α and T_β . In particular neither acquires a meridian component when transported into a Weinstein neighborhood.

Proof. The proof takes place in the already smooth target R_{aud} . By Corollary 4.3, the marked homeomorphism H transports the product collars and their push-offs from the PL model into that target. No smooth structure is placed on the source complex. A framing push-off below means the isotopy class, in $\nu(T_i) \setminus T_i$, of a sufficiently small copy of a curve on T_i , as in Definition 6.2.

Relative normal form. On an annular fiber collar write

$$\Omega = f(\theta, t) dt \wedge d\theta, \quad f > 0,$$

with core $t = 0$. Put

$$g(\theta, t) = \int_0^t (f(\theta, u) - 1) du, \quad H_s(\theta, t) = t + sg(\theta, t).$$

Then $\partial_t H_s = (1 - s) + sf > 0$. On one collar common to the compact core, H_s therefore has a smooth inverse in the radial coordinate. If $H_s(\theta, T_s(\theta, t_0)) = t_0$, then $T_0 = t_0$, $T_s(\theta, 0) = 0$, and

$$\partial_s T_s = -\frac{g(\theta, T_s)}{(1 - s) + sf(\theta, T_s)}.$$

Thus $(\theta, t_0) \mapsto (\theta, T_s)$ is the relative Moser isotopy, obtained by monotone inversion rather than an abstract flow theorem. At $s = 1$, differentiation in t_0 gives $f(\theta, T_1)\partial_{t_0}T_1 = 1$, hence $T_1^*\Omega = dt_0 \wedge d\theta$. The exact identities, fixed-core condition, and a uniform collar bound have a finite symbolic replay listed in the supplement. Apply this construction independently on the disjoint protected c and e collars. It therefore standardizes both neighborhoods simultaneously while fixing both torus cores and leaving the form unchanged outside slightly larger disjoint collars.

For the alpha collar the deck involution is $\tau(\theta, t) = (\theta + \pi, t)$ and the quotient coordinate is $\bar{\theta} = 2\theta$; write π_α for the quotient map. The preceding radial map is defined directly upstairs by replacing $\bar{\theta}$ with 2θ in its coefficient. Periodicity makes it τ -equivariant, so no covering-lift choice is involved. Moreover

$$\pi_\alpha^*(dt \wedge d\bar{\theta}) = 2 dt \wedge d\theta = d(2t) \wedge d\theta,$$

which is the required factor-two normalization. The corresponding equivariance identities are replayed in the supplement's framing inventory.

The two charts. Near T_β , ψ_0 is the identity near e , and the preceding normalization gives

$$\omega = dt \wedge d\theta_1 + K du \wedge d\theta_2, \quad K > 0;$$

the cotangent momenta are (t, Ku) . The in-fiber and base-normal push-offs are therefore constant-momentum copies. Near T_α , the mapping-torus seam is

$$(\theta_1, t, 2\pi, u) \sim (\theta_1 + \pi, t, 0, u).$$

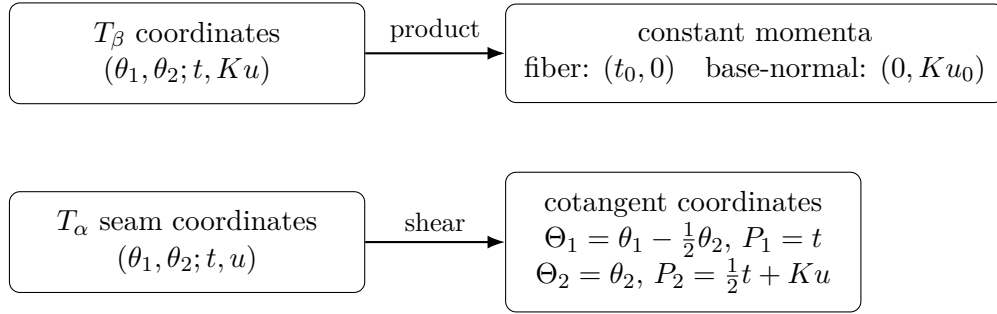
On its quotient put

$$\Theta_1 = \theta_1 - \frac{1}{2}\theta_2, \quad \Theta_2 = \theta_2, \quad P_1 = t, \quad P_2 = \frac{1}{2}t + Ku.$$

These functions respect the seam and satisfy

$$dP_1 \wedge d\Theta_1 + dP_2 \wedge d\Theta_2 = dt \wedge d\theta_1 + K du \wedge d\theta_2.$$

The fiber push-off has momentum $(t_0, t_0/2)$ and the half-drifting base push-off has momentum $(0, Ku_0)$, both constant. Hence neither acquires a meridian component, which is precisely the equality of framings used in (1). The Liouville primitive $t d\theta_1 + Ku d\theta_2$ forces these momenta by pairing with the Θ -frame; the supplement records an independent replay of that derivation, the seam, exterior algebra, and a second opposite-shear chart.



$$\begin{aligned} \text{fiber push-off: } (P_1, P_2) &= (t_0, t_0/2) \\ \text{half-drift push-off: } (P_1, P_2) &= (0, Ku_0) \end{aligned}$$

Figure 5: The two local Weinstein-coordinate calculations. Product push-offs become constant-covector copies in both charts, so the path from a push-off to the torus never winds in a normal circle. The chart-independence argument below shows that this boundary framing class is intrinsic.

Here is the boundary-class assertion in coordinates. Let γ be either named curve, let η complete it to an oriented basis of the zero-section torus, and let μ be the positively oriented circle in a cotangent fiber. After radial projection to the unit-covector boundary, a constant nonzero covector p_0 gives

$$s \longmapsto (\gamma(s), p_0/\|p_0\|).$$

Its normal-circle coordinate is constant, so its class in the boundary basis (γ, η, μ) has μ -coefficient zero. This is the precise sense in which “constant covector” excludes a meridional component. The seam map and the quotient map above preserve the displayed orientations: their pullbacks give the positive forms written there, and the chosen positive normal circle is the boundary orientation of the ordered momentum plane.

Independence of the Weinstein chart. It remains to show that the constant-momentum conclusion is a framing class rather than a feature of the displayed coordinates. Given two Lagrangian-neighborhood charts, first compose with a cotangent lift so that their restrictions to the zero section agree. Their transition is then a symplectomorphism germ $\Phi : (T^*T, T) \rightarrow (T^*T, T)$ fixing the zero section pointwise. With fiber dilation $\delta_r(q, p) = (q, rp)$, define

$$\Phi_r = \delta_r^{-1} \Phi \delta_r \quad (0 < r \leq 1).$$

Although $\delta_r^* \omega = r\omega$, the two factors cancel, so every Φ_r is symplectic. Along the zero section the symplectic condition gives

$$d\Phi = \begin{pmatrix} I & A \\ 0 & I \end{pmatrix}, \quad A = A^T.$$

Taylor expansion shows that Φ_r extends at $r = 0$ by the identity. Hence Φ_r is an isotopy of germs, relative to T , from the identity to the chart transition. To make the punctured-neighborhood assertion uniform, choose a compact germ domain over the entire named curve and then choose one nonzero constant covector copy small enough that every Φ_r is defined on it. Each Φ_r fixes the zero section and is injective, so no point of that copy can meet the zero section. Compactness of the copy times $[0, 1]$ then gives a positive distance from the zero section throughout the isotopy. Thus the whole isotopy lies in one $\nu(T) \setminus T$ and proves that the two push-offs have the same boundary class. The first-jet symplectic calculation is replayed by the checker listed in the supplement. The potentially dangerous momentum shift by a closed one-form could alter the pushed-off boundary class, but it moves the zero section unless the form vanishes and is therefore excluded here. The Lagrangian-neighborhood statement in the first paragraph of [5, §2.1], applied to each compact torus, supplies the required zero-section Weinstein charts; the argument above supplies the precise chart-independence consequence needed here. $\square$

Completion of the proof of Theorem 2.2. Lemma 6.1 makes the two disjoint audit tori Lagrangian, and Theorem 6.3 identifies the two product longitudes in Definition 2.1 with their Lagrangian-framing classes. The two unit-coefficient fillings are therefore Luttinger surgeries in the sense of [5, Definition 2.1]; the symplectic form on the filled manifold is the one asserted in [5, Proposition 2.2]. That construction supplies a symplectic form after each surgery which agrees with ω_K outside the chosen torus neighborhood. Performing the surgeries successively gives a symplectic form on V_{aud} . The fiber F chosen in the proof above is disjoint from both surgery neighborhoods, so the resulting form still restricts positively to F . $\square$

7 The audit double and the exotic $S^2 \times S^2$

This section proves Theorem 2.3. The boundary of V_{aud} is the mapping torus of the boundary monodromy. The first subsection defines an involution of it from the clutching map φ_0 alone and records the topology of the double. The second shows that the symplectic form of Lemma 6.1 descends to the double, so that the four surgeries are Luttinger surgeries on a closed symplectic surface bundle. The third lists every external result used, with the place where its hypotheses are discharged, and assembles the proof.

7.1 The intrinsic boundary involution

Throughout this section φ_0 and ψ_0 are the smooth representatives of Lemma 6.1 on the smooth surface F , and R_{aud} is the smooth bundle they clutch (Corollary 4.3). Write $h: F \rightarrow F$ for the monodromy of the boundary bundle $\partial R_{\text{aud}} \rightarrow \partial \Sigma_{1,1}$, the commutator of φ_0 and ψ_0 determined by the oriented boundary word, a diffeomorphism, and

$$M_h = (F \times [0, 1]) / ((z, 1) \sim (h(z), 0))$$

for its mapping torus, a closed smooth 3-manifold with the base coordinate $t \in \mathbb{R}/\mathbb{Z}$, so that $\partial R_{\text{aud}} = M_h$. Since both surgeries take place in the interior, $\partial V_{\text{aud}} = M_h$ as well.

Lemma 7.1 (the audit boundary involution). Define

$$\sigma_{\text{aud}}: M_h \rightarrow M_h, \quad \sigma_{\text{aud}}[z, t] = [\varphi_0(z), 1 - t].$$

Then:

- (i) σ_{aud} is well defined, and is a fiber-preserving smooth involution of M_h covering the reflection $t \mapsto 1 - t$ of $\partial \Sigma_{1,1}$;
- (ii) σ_{aud} reverses the orientation of ∂R_{aud} ;
- (iii) σ_{aud} carries each of the circles $\{p\} \times S^1$ and $\{O\} \times S^1$, the boundaries of the two fixed-point sections, onto itself;
- (iv) σ_{aud} fixes $[p, \frac{1}{2}]$, $[O, \frac{1}{2}]$, $[p, 0]$, and $[O, 0]$; in particular it is not free.

No property of ψ_0 beyond its being a diffeomorphism fixing p and O is used, and no property of φ_0 beyond its being an orientation-preserving smooth involution fixing p and O .

Proof. (i) The only identification in M_h is $(z, 1) \sim (h(z), 0)$. Under the displayed formula, $(z, 1) \mapsto (\varphi_0 z, 0)$ while $(hz, 0) \mapsto (\varphi_0 hz, 1) \sim (h\varphi_0 hz, 0)$. These agree for every z if and only if $\varphi_0 = h\varphi_0 h$ as maps, that is, using $\varphi_0^{-1} = \varphi_0$, if and only if

$$\varphi_0 h \varphi_0^{-1} = h^{-1}. \quad (2)$$

Now h is a commutator of φ_0 and ψ_0 . For the ordering $h = \varphi_0 \psi_0 \varphi_0^{-1} \psi_0^{-1}$,

$$\varphi_0 h \varphi_0^{-1} = \varphi_0^2 \psi_0 \varphi_0^{-1} \psi_0^{-1} \varphi_0^{-1} = \psi_0 \varphi_0 \psi_0^{-1} \varphi_0 = (\varphi_0^{-1} \psi_0 \varphi_0 \psi_0^{-1})^{-1} = h^{-1},$$

using only $\varphi_0^2 = \text{id}$; the other three orderings and orientations of the commutator are checked in the same way (the machine control `luttinger/sigma_aud/sigma_aud_check.py` verifies all four exactly on the based generators of $\pi_1(F, p)$, using the certified monodromy actions). So σ_{aud} is well defined. It is smooth: away from the seam it is $\varphi_0 \times (t \mapsto 1 - t)$; in the seam chart $F \times (1 - \varepsilon, 1 + \varepsilon)$ of M_h , in which (w, s) with $s > 1$ denotes $(h(w), s - 1)$, it reads $(w, s) \mapsto (h^{-1}\varphi_0 w, 2 - s)$ for $s < 1$ and $(w, s) \mapsto (\varphi_0 h w, 2 - s)$ for $s > 1$, and these are one smooth map precisely because (2) holds as an identity of maps. Being an involution it is a diffeomorphism. (On the PL model it is PL after subdividing $[0, 1]$ symmetrically about $\frac{1}{2}$; only the smooth statement is used below.) It preserves fibers and covers $t \mapsto 1 - t$ by construction, and $\sigma_{\text{aud}}^2[z, t] = [\varphi_0^2 z, t] = [z, t]$.

(ii) Orient M_h as $F \times [0, 1]$, fiber first. By Definition 2.1, φ_0 preserves the orientation of F ; the reflection reverses that of $[0, 1]$; the product reverses.

(iii) $\varphi_0(p) = p$ and $\varphi_0(O) = O$.

(iv) A point $[z, t]$ is fixed only if $t = 1 - t$ in $\mathbb{R}/\mathbb{Z}$, so $t = \frac{1}{2}$ or $t = 0$. Over $t = \frac{1}{2}$ the fixed points are $\{z : \varphi_0 z = z\} = \{p, O\}$. Over the seam, $\sigma_{\text{aud}}[z, 0] = [\varphi_0 z, 1] = [h\varphi_0 z, 0]$, so the fixed points are those of $h\varphi_0$; since φ_0 and ψ_0 both fix p and O , so does h , and p, O are among them. (By (2), $(h\varphi_0)^2 = h\varphi_0 h\varphi_0 = \text{id}$, so $h\varphi_0$ is itself an orientation-preserving involution of F ; its full fixed set is finite but is not needed.) $\square$

Remark 7.2. σ_{aud} is not free. Freeness of a boundary involution is needed only to form a quotient $W = V/\sigma$, as in the surgery scheme of [4]; nothing in the proof of Theorem 2.3 uses it, and the quotient and everything downstream of it are outside the scope of this article. The supplement's historical transfer chain imports a free involution through the attribution checklist; this article uses σ_{aud} only.

Corollary 7.3 (the audit double). Let $\hat{\Sigma} = \Sigma_{1,1} \cup_{\rho} \Sigma_{1,1}$ be the double of the base along the reflection $\rho(t) = 1 - t$ of its boundary circle, a closed oriented surface of genus two, and put

$$\hat{R} = R_{\text{aud}} \cup_{\sigma_{\text{aud}}} R_{\text{aud}}, \quad Z_{\text{aud}} = V_{\text{aud}} \cup_{\sigma_{\text{aud}}} V_{\text{aud}},$$

both copies carrying the orientation of R_{aud} , glued along the boundary diffeomorphism σ_{aud} of Lemma 7.1, with the smooth structure on the union fixed by the collar extension of Lemma 7.4(ii). Then:

- (a) $\hat{R}$ is a closed oriented smooth genus-two surface bundle over $\hat{\Sigma}$, $\hat{\Sigma}$ has genus two, and Z_{aud} is a closed oriented smooth 4-manifold;
- (b) Z_{aud} is obtained from $\hat{R}$ by the two audit surgeries and their images under the swap of the two copies, performed on four pairwise disjoint tori in the interior of $\hat{R}$, all disjoint from a fiber F over a base point away from α, β and their mirrors, and from the doubled section $\hat{\Gamma} = \Gamma \cup_{\sigma_{\text{aud}}} \Gamma$;
- (c) $\hat{\Gamma}$ is a closed orientable surface of genus two with $\hat{\Gamma} \cdot \hat{\Gamma} = 0$, $F \cdot F = 0$, and $F \cdot \hat{\Gamma} = 1$.

Proof. (a) Gluing two copies of a smooth manifold with boundary along a boundary diffeomorphism, with collars chosen as in Lemma 7.4(ii), produces a closed smooth manifold, and any two collar choices give diffeomorphic results; this is how $\hat{R}$ and Z_{aud} are smooth manifolds, and V_{aud} differs from R_{aud} only in its interior. Gluing two copies of a smooth bundle by a fiber-preserving diffeomorphism covering a diffeomorphism of the base boundary produces a smooth bundle over the glued base; here the base gluing is ρ , and $\Sigma_{1,1} \cup_{\rho} \Sigma_{1,1}$ is the double of a once-punctured torus, hence closed, orientable, of genus two. Both copies of R_{aud} induce the same orientation on the common boundary, and by Lemma 7.1(ii) the gluing map reverses it, so $\hat{R}$ is oriented compatibly with both copies. (Factorwise: the base gluing ρ reverses the boundary circle, so $\hat{\Sigma}$ restricts to the given orientation on each copy of $\Sigma_{1,1}$; the fiber gluing φ_0 preserves the orientation of F ; so neither factor of the product orientation is reversed on the second copy.)

(b) T_{α} and T_{β} lie over the spine loops α, β , which are interior to $\Sigma_{1,1}$, so both tori and their product collars miss ∂R_{aud} ; the surgeries therefore commute with the gluing, and their images under the swap are surgeries on the mirror tori. The four tori are disjoint: the two audit tori are disjoint by construction and lie in one copy, their mirrors in the other, and the two copies meet only along ∂R_{aud} , which none of them meets. A fiber over a base point away from the four loops misses all four tori. The section Γ through p misses T_{α} and T_{β} because p lies on neither c nor e ; the same holds in the mirror copy; and by Lemma 7.1(iii) the two halves glue along $\partial\Gamma = \{p\} \times S^1$ to a closed surface $\hat{\Gamma}$.

(c) $\hat{\Gamma}$ is the double of the once-punctured torus Γ along its boundary circle, hence closed, orientable, of genus two. For the self-intersection, $\hat{\Gamma} \cdot \hat{\Gamma}$ is the Euler number of the oriented normal bundle $\nu(\hat{\Gamma})$. Over each copy, $\Gamma = \{p\} \times \Sigma_{1,1}$ is horizontal in the flat product atlas of Lemma 6.1, so $\nu(\Gamma)$ is the flat oriented plane bundle with fiber $T_p F$ and

holonomy $d\varphi_0(p)$ along α and $d\psi_0(p) = I$ along β (ψ_0 is the identity near p). Such a bundle is trivial, its base being homotopy equivalent to a graph, and its holonomy around $\partial\Gamma$ is the commutator $[d\varphi_0(p), d\psi_0(p)] = I$, so the flat structure frames it along $\partial\Gamma$; that flat boundary framing has degree zero relative to a global trivialization, because with respect to an invariant inner product the connection form A is a closed real 1-form and the rotation is $\frac{1}{2\pi} \oint_{\partial\Gamma} A = \frac{1}{2\pi} \int_{\Gamma} dA = 0$. Along $\partial\Gamma = \{p\} \times S^1$ the gluing map σ_{aud} acts on $T_p F$ by the constant map $d\varphi_0(p) = -I$ (φ_0 is a smooth involution with p an isolated fixed point, so it is linearizable at p with derivative $-I$), which has degree zero in $SO(2)$. Hence the clutching map of $\nu(\hat{\Gamma})$ between the two flat boundary framings has degree zero and $\hat{\Gamma} \cdot \hat{\Gamma} = 0$. The certificate `luttinger/pl_self_intersection.py` confirms this combinatorially: it constructs $\hat{\Gamma}$ as a simplicial surface with f -vector $(138, 420, 280)$, on whose 24-cycle link of p in the fiber triangulation φ_0 acts by the half-turn (shift 12), i.e. by $-I$; it builds the triangulated normal disk bundle of $\hat{\Gamma}$ for each of the four constant boundary clutching rotations, including this one (its “constant clutch shift 2”), and in each case exhibits a disjoint simplicial push-off together with the radial PL cylinder joining it to the zero section, so the signed intersection chain is empty. A fiber has trivial normal bundle, so $F \cdot F = 0$, and a fiber meets a section transversally in one point, so $F \cdot \hat{\Gamma} = 1$. $\square$

7.2 The symplectic form on the audit double

Keep the notation of Lemma 7.1 and Corollary 7.3: $\hat{\Sigma} = \Sigma_{1,1} \cup_{\rho} \Sigma_{1,1}$, $\hat{R} = R_{\text{aud}} \cup_{\sigma_{\text{aud}}} R_{\text{aud}}$, $Z_{\text{aud}} = V_{\text{aud}} \cup_{\sigma_{\text{aud}}} V_{\text{aud}}$, with $\hat{\pi}: \hat{R} \rightarrow \hat{\Sigma}$ the projection and $F, \hat{\Gamma}$ the fiber and doubled section of that corollary.

Lemma 7.4 (symplectic form on the audit double). Let Ω be the positive area form on F of Lemma 6.1, preserved by φ_0 and ψ_0 . Choose the positive area form η on $\Sigma_{1,1}$ so that on a collar $\partial\Sigma_{1,1} \times (-\varepsilon, 0]$ with boundary parameter $t \in \mathbb{R}/\mathbb{Z}$ and outward-increasing collar coordinate s it equals $ds \wedge dt$, and so that the collar is disjoint from the annular collars of α, β used in that lemma. Then:

- (i) η on each copy glues to a smooth positive area form $\hat{\eta}$ on $\hat{\Sigma}$;
- (ii) the vertical form Ω_{vert} on each copy glues to a smooth closed vertical form $\hat{\Omega}_{\text{vert}}$ on $\hat{R}$;
- (iii) for every $K > 0$, $\hat{\omega}_K = \hat{\Omega}_{\text{vert}} + K\hat{\pi}^*\hat{\eta}$ is a symplectic form on $\hat{R}$ restricting to ω_K on each copy;
- (iv) $\hat{\omega}_K$ is positive on F and on $\hat{\Gamma}$;
- (v) the four surgery tori of Corollary 7.3(b) are Lagrangian for $\hat{\omega}_K$, and each has a protected neighborhood with the split coordinates of Lemma 6.1, disjoint from the seam collar $\hat{\pi}^{-1}(\partial\Sigma_{1,1} \times (-\varepsilon, \varepsilon))$.

Proof. (i) On the seam collar of $\hat{\Sigma}$ use the chart $(t, s) \in \mathbb{R}/\mathbb{Z} \times (-\varepsilon, \varepsilon)$ in which $s \leq 0$ is the first copy’s collar and, for $s \geq 0$, the point (t, s) is the second copy’s collar point with its own coordinates $t' = \rho(t) = 1 - t$ and $s' = -s$: the second copy’s outward direction is toward the seam, so its outward-increasing coordinate is $-s$. In its own coordinates the second copy’s form is $ds' \wedge dt' = d(-s) \wedge d(1 - t) = ds \wedge dt$. So the two copies of η are the single form $ds \wedge dt$

on the whole chart, and $\hat{\eta}$ is smooth. It is positive on both copies since each is oriented as $\Sigma_{1,1}$ and $\hat{\Sigma}$ restricts to those orientations (Corollary 7.3(a)).

(ii) Over the seam collar the flat product atlas of Lemma 6.1 presents the bundle as $M_h \times (-\varepsilon, 0]$ with Ω_{vert} equal to Ω on every fiber; this is well defined because the seam map h of M_h is a product of the clutching maps and so preserves Ω . Extend the gluing map over the two collars as $(z, t, s) \mapsto (\varphi_0 z, 1 - t, -s)$; this fixes the smooth structure of $\hat{R}$ near the seam (any two collar choices give diffeomorphic doubles), and in the flat atlas it is, on every fiber, the single map φ_0 , which is locally constant in the base and preserves Ω . Hence the transition maps of the flat atlas of $\hat{R}$ obtained by adjoining σ_{aud} all lie in the group generated by φ_0 and ψ_0 , all preserve Ω , and the fiberwise form Ω descends to a globally defined vertical form $\hat{\Omega}_{\text{vert}}$ on $\hat{R}$. It is closed because in each flat product chart it is the pullback of the closed form Ω from the fiber factor.

(iii) $d\hat{\omega}_K = 0$ by (i) and (ii), and with the fiber and base orientations fixed by Ω and $\hat{\eta}$, $\hat{\omega}_K^2 = 2K \hat{\Omega}_{\text{vert}} \wedge \hat{\pi}^* \hat{\eta} > 0$, exactly as in Lemma 6.1. On each copy the form is $\Omega_{\text{vert}} + K \pi^* \eta = \omega_K$ by construction.

(iv) F is a fiber, on which $\hat{\omega}_K$ restricts to $\Omega > 0$. $\hat{\Gamma}$ is the doubled section through the common fixed point p ; in every flat product chart it is $\{p\} \times U$, so $\hat{\Omega}_{\text{vert}}$ restricts to zero on it and $\hat{\omega}_K|_{\hat{\Gamma}} = K\hat{\eta} > 0$.

(v) Each of the four tori lies in the interior of one copy, over the annular collar of α or β in that copy, which is disjoint from the seam collar by the choice of η . On that copy $\hat{\omega}_K = \omega_K$, so Lemma 6.1 applies verbatim: the torus has one fiber tangent and one base tangent, both summands vanish on it, and the protected split coordinates are the ones displayed there. The mirror tori are the images of T_α, T_β under the swap of the two copies, which is an isomorphism of the second copy with R_{aud} carrying $\hat{\omega}_K|_{\text{copy 2}}$ to ω_K . $\square$

Corollary 7.5 (Z_{aud} is symplectic). Z_{aud} carries a symplectic form $\hat{\omega}$ that agrees with $\hat{\omega}_K$ outside the four surgery neighborhoods, restricts positively to F and to $\hat{\Gamma}$, and is obtained from $(\hat{R}, \hat{\omega}_K)$ by four Luttinger surgeries in the sense of [5, Definition 2.1].

Proof. By Lemma 7.4(v) and Theorem 6.3 applied in each copy, the four tori are Lagrangian and the four unit-coefficient filling slopes are their Lagrangian-framing classes; the surgeries of Corollary 7.3(b) are therefore Luttinger surgeries, and [5, Proposition 2.2] supplies, one surgery at a time, a symplectic form agreeing with the previous one outside the surgered neighborhood. The four neighborhoods are disjoint from one another, from F , and from $\hat{\Gamma}$ (Corollary 7.3(b)), so the final form still restricts positively to F and $\hat{\Gamma}$ by Lemma 7.4(iv). $\square$

7.3 Hypotheses of the external results, discharged

The proof of Theorem 2.3 uses six results from the literature. Each is stated with its hypotheses, and each hypothesis is discharged by a statement about the audit object proved in this article. No comparison with any other author's construction enters.

Van Kampen. $Z_{\text{aud}} = V_{\text{aud}} \cup_{\partial} V_{\text{aud}}$ with connected, bicollared common boundary $\partial V_{\text{aud}} = M_h$ and $\pi_1(V_{\text{aud}}) = 1$ (Theorem 2.2); hence $\pi_1(Z_{\text{aud}}) = 1$.

Freedman [8]. Closed simply connected topological 4-manifolds are classified by intersection form and Kirby–Siebenmann invariant, the latter determined by $\sigma/8$ when the form is

even. Hypotheses discharged in Proposition 7.6: simple connectivity, $b_2 = 2$, form H on the basis $(F, \hat{\Gamma})$, smoothness (so Kirby–Siebenmann vanishes).

Thurston [7]. A closed surface bundle over a surface with homologically essential fiber is symplectic with symplectic fibers. We do not invoke it: Lemma 7.4 constructs the form directly (the fiber is essential anyway, since $F \cdot \hat{\Gamma} = 1$).

Auroux–Donaldson–Katzarkov [5] (Luttinger [6]). Luttinger surgery on a Lagrangian torus along its Lagrangian framing yields a symplectic manifold with a form agreeing with the old one away from the torus. Hypotheses discharged in Lemma 7.4(v) and Theorem 6.3: Lagrangian tori, Lagrangian-framed slopes.

Ho–Li [9, Theorem 1.1]. Luttinger surgery preserves the symplectic Kodaira dimension. The surgery there is the one of [9, Section 2.1], which follows [5], performed on a Lagrangian torus of a closed symplectic 4-manifold. Hypotheses discharged in Corollary 7.5: each of the four surgeries is a Luttinger surgery in the sense of [5, Definition 2.1] on a Lagrangian torus of the closed symplectic manifold produced by the previous one.

Liu [12], as stated in [11, Theorem 2.4(2)]. A minimal closed symplectic 4-manifold has Kodaira dimension $-\infty$ only if it is rational or ruled, where rational and ruled are defined by the underlying smooth manifold $(S^2 \times S^2, \mathbb{CP}^2 \# k \overline{\mathbb{CP}}^2)$, or an S^2 -bundle over a surface blown up $k \geq 0$ times). Hypothesis: minimality, discharged for $(\hat{R}, \hat{\omega}_K)$ in Step 1 of the proof below by asphericity.

Li [10], [11, p. 251]. The Kodaira dimension of a closed symplectic 4-manifold depends only on the oriented diffeomorphism type: two symplectic forms on one closed oriented 4-manifold, both compatible with its orientation, have the same κ . Hypothesis: a common orientation, arranged in Step 3 of the proof below.

Proposition 7.6 (topology of Z_{aud}). Z_{aud} is closed, simply connected, and spin, with $b_2(Z_{\text{aud}}) = 2$, signature 0, and intersection form H on the basis $(F, \hat{\Gamma})$. Consequently Z_{aud} is homeomorphic to $S^2 \times S^2$.

Proof. Closed: ∂V_{aud} is glued to itself. Simply connected by van Kampen as above. $\chi(Z_{\text{aud}}) = 2\chi(V_{\text{aud}}) - \chi(M_h) = 4 - 0 = 4$ (the proof of Theorem 2.2 gives $\chi(V_{\text{aud}}) = 2$; a mapping torus has Euler characteristic 0), so $b_2 = 2$ with H_2 torsion-free. By Corollary 7.3(c), $F \cdot F = 0$, $\hat{\Gamma} \cdot \hat{\Gamma} = 0$, $F \cdot \hat{\Gamma} = 1$: the Gram matrix of $(F, \hat{\Gamma})$ is H , of determinant -1 , hence unimodular; a unimodular rank-two sublattice of the unimodular rank-two lattice $H_2(Z_{\text{aud}})$ has index one, so $(F, \hat{\Gamma})$ is a basis and the form is H . H is even, so $w_2 = 0$ by Wu’s formula and Z_{aud} is spin; its signature is 0. Z_{aud} is smooth, so its Kirby–Siebenmann invariant vanishes, and Freedman’s classification gives $Z_{\text{aud}} \cong S^2 \times S^2$. $\square$

Proof of Theorem 2.3. Symplectic and homeomorphic to $S^2 \times S^2$ by Corollary 7.5 and Proposition 7.6. For non-diffeomorphism we use the symplectic Kodaira dimension κ of [10]: for a minimal closed symplectic 4-manifold (M, ω) with canonical class K_ω , $\kappa(M, \omega) = -\infty$ if and only if $K_\omega \cdot [\omega] < 0$ or $K_\omega^2 < 0$ [11, Definition 2.2]. Throughout, a manifold is oriented and its symplectic forms are compatible with that orientation.

Step 1: $\kappa(\hat{R}, \hat{\omega}_K) \neq -\infty$. $\hat{R}$ is a surface bundle with aspherical fiber and base, hence aspherical. A symplectic 4-manifold with $\pi_2 = 0$ is minimal (an exceptional sphere would be null-homotopic, hence null-homologous, of square 0, not -1) and is neither rational nor ruled: each of those contains an embedded sphere that is homologically nonzero (a factor sphere, a line, an exceptional sphere, or a fiber of an S^2 -bundle, which meets a section once), so has $\pi_2 \neq 0$. By Liu's theorem [12], [11, Theorem 2.4(2)], a minimal symplectic 4-manifold with $\kappa = -\infty$ is rational or ruled. Hence $\kappa(\hat{R}, \hat{\omega}_K) \neq -\infty$.

Step 2: $\kappa(Z_{\text{aud}}, \hat{\omega}) \neq -\infty$. By Corollary 7.5, $(Z_{\text{aud}}, \hat{\omega})$ is obtained from $(\hat{R}, \hat{\omega}_K)$ by four successive Luttinger surgeries, each on a Lagrangian torus of the closed symplectic 4-manifold produced by the previous one. Each preserves κ [9, Theorem 1.1].

Step 3: a pulled-back product form. Suppose $\psi: Z_{\text{aud}} \rightarrow S^2 \times S^2$ were a diffeomorphism. Orient Z_{aud} by $\hat{\omega}$ and $S^2 \times S^2$ as a product of oriented spheres. If ψ reverses orientation, replace it by $(r \times \text{id}) \circ \psi$ with r a reflection of S^2 ; so ψ may be taken orientation-preserving. Let $\omega_{a,b} = a \text{pr}_1^* \sigma + b \text{pr}_2^* \sigma$ ($a, b > 0$, σ an area form of total area 1 on S^2), a symplectic form on $S^2 \times S^2$ compatible with its orientation. Then $\psi^* \omega_{a,b}$ is a symplectic form on Z_{aud} compatible with the $\hat{\omega}$ -orientation, and $\kappa(Z_{\text{aud}}, \psi^* \omega_{a,b}) = \kappa(S^2 \times S^2, \omega_{a,b})$ since ψ is a symplectomorphism. The latter is $-\infty$ by the definition: the intersection form of $S^2 \times S^2$ is even, so no class has square -1 and the manifold is minimal; and the canonical class of the product form evaluates to $-\chi(S^2) = -2$ on each factor sphere, so $K \cdot [\omega_{a,b}] = -2a - 2b < 0$.

Step 4. $\hat{\omega}$ and $\psi^* \omega_{a,b}$ are symplectic forms on Z_{aud} compatible with one orientation, so $\kappa(Z_{\text{aud}}, \hat{\omega}) = \kappa(Z_{\text{aud}}, \psi^* \omega_{a,b})$ by [10], [11, p. 251]: the Kodaira dimension depends only on the oriented diffeomorphism type. Steps 2 and 3 make the two sides $\neq -\infty$ and $= -\infty$. Hence no such ψ exists, and Z_{aud} is not diffeomorphic to $S^2 \times S^2$. $\square$

Remark 7.7 (what the theorem does and does not say). Theorem 2.3 is a statement about the object of Definition 2.1 and its double along σ_{aud} . It does not assert that Z_{aud} is the manifold proposed by Wuebben [1]; that attribution question is the subject of Section 8, and no statement there enters this proof.

8 Attribution: relation to Wuebben's construction, and review boundary

Nothing in Theorem 2.2 or Theorem 2.3 identifies V_{aud} with Wuebben's fixed surgery member or Z_{aud} with his double. The companion supplement records fourteen comparison clauses D1–D14: textual and diagrammatic clauses D1–D11 specify the proposed dictionary, while D12–D14 ask for the actual relative smooth identification of the two Lagrangian tori, framing conventions, and marked bundles. Those clauses are a checklist for attribution, not hypotheses of any theorem in this article. Even if every clause held, Z_{aud} is glued by σ_{aud} rather than by the free involution of [4], so the two doubles would still require a separate comparison. The exotic-manifold conclusion asserted here is Theorem 2.3, about Z_{aud} .

8.1 The reported contrary computation and live failure locations

Wuebben's public repository reports that Lidman and Piccirillo obtained a fundamental-group calculation disagreeing with his, currently summarized as $\pi_1(V) \neq 1$. Their published paper [4] proves $H_1(V) = 0$ and normal generation by $\pi_1(F)$ but does not publish this contrary

presentation. No relation sheet, generator dictionary, or nontriviality witness for the reported calculation is available in the materials examined here.

The finite Theorem 3.1 is unaffected because its objects are explicit strings, and Theorem 2.3 is a statement about the explicit object Z_{aud} . A contrary calculation about a different member says nothing about either; one about the same member would locate an error in one of the two computations. If the contrary calculation concerns the same geometric member, the live explanations are deliberately left neutral: a source-to-audit clause may fail; a displayed geometric identification or finite checker may be defective; the contrary calculation may be defective; or the computations may concern different members or conventions. A relation-by-relation reconciliation requires the missing presentation, dictionary, and witness. Until that material exists, this article makes no claim about Wuebben’s V .

8.2 Independent review targets

Three kinds of independent review remain valuable. First, both theorem-critical filled-group checkers arose within the same AI-assisted development process. Separate languages, negative controls, and the mathematical specification reduce ordinary software risk, but do not replace an external conformance audit of one checker.

Second, the principal geometric risk for simple connectivity lies in Sections 4–5, not in the certificate arithmetic. A PL reviewer should check the equivariant marked-fiber extension, relative clutching, simplex-link local-flatness conversion, normal-frontier equivalence, both commonly based peripheral words, and the claim that the drilled transport homotopy stays in the complement. Those are the steps that turn the finite presentation into $\pi_1(V_{\text{aud}})$.

Third, a symplectic reviewer should check that the split form in Lemma 6.1 descends to the double as claimed in Lemma 7.4, that the seam and normal-circle orientations agree with the filling basis, that constant-covector copies have zero meridional coefficient, and that the chart-independence isotopy remains in a uniform punctured neighborhood. The finite checks localize these questions; they do not replace specialist review. The supplement gives exact files and a longer review checklist.

The 100-case framing scan is not evidence for the dictionary: every sampled shifted presentation was reported trivial, as is every fiber-direction shift of the filling words that has been enumerated to completion (Remark 5.5). Proposition 5.8 supplies the one transparent negative control: infinitely many non-unit slope fillings have nontrivial abelianization.

9 Conclusion

Ten relations among eight based loops in the torus complement, with the two product-filling words, present the trivial group; the geometric content of simple connectivity is that the relations, the generation statement, and the filling words hold for the explicitly defined V_{aud} (Section 5.1). Four manifest-bound presentations, derived from the triangulation with no hand-written word, are trivial under a complete certificate specification and confirm the same conclusion by a second route; the geometric reconstruction identifies one of them with $\pi_1(V_{\text{aud}})$, yielding the intrinsic topological statements of Theorem 2.2; the framing theorem supplies its symplectic structure. Doubling along the intrinsic boundary involution σ_{aud} and descending that form gives Theorem 2.3: Z_{aud} is a closed symplectic four-manifold homeomorphic but not diffeomorphic to $S^2 \times S^2$. The separation between finite derivation, geometric conversion,

symplectic framing, and doubling is the article’s contribution; the final classification and non-diffeomorphism deductions use named external theorems whose hypotheses are discharged in the text.

Whether Wuebben’s text and diagrams specify the same marked smooth surgery data, and whether his double coincides with Z_{aud} , remain open. The supplement records that attribution question and the historical transfer chain without promoting either into a theorem of this article.

10 Data and reproducibility

The immutable certificate set, frozen inputs, and reconstruction programs are in the public repository and its v2.4.0 archival deposit [3]:

<https://github.com/VentiMath/exotic-s2xs2-verification/releases/tag/v2.4.0>,
<https://doi.org/10.5281/zenodo.22254457>.

That version DOI identifies this immutable release. The proof manifest, verification-tree object, and existence-chain certificate printed in the supplement are the ones bound by that release; the certificate binds the doubling lemmas of Section 7 by hashing this manuscript. The concept DOI <https://doi.org/10.5281/zenodo.22169753> resolves to the newest archived version.

The supplement gives the sealed-input digest, git tree object, literal replay commands, expected outputs, negative controls, and source-extraction pins. Certificate-only replay requires standard Python or Ruby; full reconstruction uses the recorded GAP 4.11.1 and KBMAG 1.5.9 environment. The theorem about V_{aud} replays from frozen files without locally supplied copies of either source paper.

References

- [1] B. J. Wuebben, *An exotic $S^2 \times S^2$ and an exotic $\mathbb{CP}^2 \# \overline{\mathbb{CP}^2}$* , arXiv:2608.17267v1 (2026).
- [2] B. J. Wuebben, *Public verification repository accompanying An exotic $S^2 \times S^2$ and an exotic $\mathbb{CP}^2 \# \overline{\mathbb{CP}^2}$* , commit 82fbf7ef4c2f75c75ff8e4239af8c258f244cdbc (2026); see `scripts/pi1_grid.g`, `logs/pi1_grid.log`, `scripts/vsens.g`, and `logs/vsens_out.txt`, <https://github.com/bwuebben/exotic-s2xs2>.
- [3] J. Clyde, *An exotic $S^2 \times S^2$ from a certificate-checked surface-bundle surgery manifold*, verification artifact v2.4.0, Zenodo (2026), <https://doi.org/10.5281/zenodo.22254457>.
- [4] T. Lidman and L. Piccirillo, *Distinguishing closed 4-manifolds by slicing*, arXiv:2505.14387v1 (2025).
- [5] D. Auroux, S. K. Donaldson, and L. Katzarkov, *Luttinger surgery along Lagrangian tori and non-isotopy for singular symplectic plane curves*, Math. Ann. **326** (2003), 185–203.
- [6] K. M. Luttinger, *Lagrangian tori in $\mathbb{R}^4$* , J. Differential Geom. **42** (1995), 220–228.
- [7] W. P. Thurston, *Some simple examples of symplectic manifolds*, Proc. Amer. Math. Soc. **55** (1976), 467–468.
- [8] M. H. Freedman, *The topology of four-dimensional manifolds*, J. Differential Geom. **17** (1982), 357–453.
- [9] C.-I. Ho and T.-J. Li, *Luttinger surgery and Kodaira dimension*, Asian J. Math. **16** (2012), 299–318.

- [10] T.-J. Li, *Symplectic 4-manifolds with Kodaira dimension zero*, J. Differential Geom. **74** (2006), 321–352.
- [11] T.-J. Li, *The Kodaira dimension of symplectic 4-manifolds*, in: Floer homology, gauge theory, and low-dimensional topology (D. A. Ellwood, P. S. Ozsváth, A. I. Stipsicz, Z. Szabó, eds.), Clay Math. Proc. **5**, Amer. Math. Soc., Providence, RI, 2006, pp. 249–261.
- [12] A.-K. Liu, *Some new applications of general wall crossing formula, Gompf’s conjecture and its applications*, Math. Res. Lett. **3** (1996), 569–585.
- [13] A. Constantin and B. Kolev, *The theorem of Kerékjártó on periodic homeomorphisms of the disc and the sphere*, Enseign. Math. (2) **40** (1994), 193–204.
- [14] C. P. Rourke and B. J. Sanderson, *Introduction to Piecewise-Linear Topology*, Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 69, Springer-Verlag, 1972.
- [15] A. E. Hatcher, *Linearization in 3-dimensional topology*, Proceedings of the International Congress of Mathematicians (Helsinki, 1978), pp. 463–468, Academia Scientiarum Fennica, 1980.
- [16] A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002.
- [17] J. Milnor and J. Stasheff, *Characteristic Classes*, Annals of Mathematics Studies, vol. 76, Princeton University Press, 1974.